Engineering Applications of Artificial Intelligence 158 (2025) 111412

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

GOM-MMOEA: Multimodal multi-objective evolutionary algorithm based on global orchestration mechanism

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ARTICLE INFO

Keywords:
Multimodal multi-objective optimization
Diversity
Convergence
Orchestration

ABSTRACT

The core challenge of multimodal multi-objective optimization lies in identifying and discovering multiple equivalent sets of Pareto-optimal solutions, thereby offering diverse options for decision-makers. However, most existing algorithms suffer from premature convergence when tackling such problems. This issue often arises due to inadequate population diversity and ineffective global exploration mechanisms during the search process, which causes the algorithm to become trapped in local optima and hinders the exploration of other promising regions in the decision space. To address this challenge, this paper proposes a multimodal multi-objective evolutionary algorithm based on a global orchestration mechanism. First, the algorithm constructs and dynamically updates an orchestration vector to guide the search toward optimal solutions and accelerate population convergence. Second, an orchestration vector update strategy is designed to gradually diminish the influence of inferior solutions, thereby preventing convergence to local optima. During the early stages of evolution, larger increments are applied to high-quality solutions to speed up convergence, while these increments are gradually reduced over time to promote global exploration. Finally, a novel parent selection mechanism is introduced, which dynamically adjusts selection probabilities to optimize the search process while preserving population diversity. Moreover, the algorithm adopts a triple population synergistic orchestration method that simultaneously considers both the objective and decision spaces. Experimental results demonstrate that the proposed algorithm outperforms several state-of-the-art methods across a range of benchmark test problems.

1. Introduction

Multi-objective optimization problems (MOPs) are complex challenges in the field of optimization, primarily due to the need to simultaneously optimize multiple, often conflicting, objective functions (Li et al., 2021a, 2023a; Zhou et al., 2021; Zhang et al., 2020; Zheng and Sun, 2022; Jangir et al., 2023; Premkumar et al., 2022; Jangir and Trivedi, 2018). In such problems, there is not always a single solution that performs most effectively across all objectives. However, the goal of a multi-objective optimization problem is to identify a set of solutions (Kalita et al., 2024a, 2024b, 2024c, 2024d, 2024e; Ravichandran et al., 2024; Jangir et al., 2021; Premkumar et al., 2021), known as the Pareto optimal set, that offer the relatively better choice between the multiple objectives.

Mathematically, MOPs can be formulated as:

$$\begin{aligned} &min\ F(x) = \{f_1(x), f_2(x), \ldots, f_m(x)\} \\ &s.t. x = (x_1, x_2, \ldots, x_n) \in \Omega \end{aligned} \tag{1}$$

where Ω represents the decision space, m denotes the number of objective functions, and x is a decision vector consisting of n decision variables. A solution x_a is said to dominate a solution x_b if, for all $i = 1,2, \ldots, m$, the value of the solution x_a on the objective function $f(x)$ is less than or equal to the value of the solution x_b, and the value of x_a is better than x_b on at least one objective function j. A solution is Pareto optimal (PS) if it is not dominated by any other solution. And the mapping of the Pareto optimal set in the objective space, i.e., the set consisting of the objective function values of these solutions, is called the Pareto optimal front (PF).

Numerous evolutionary optimization algorithms (Agrawal et al., 2025; Jangir et al., 2024a, 2024b; Mirjalili et al., 2017a, 2017b; Kumar et al., 2021; Premkumar et al., 2020) have been proposed by researchers

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https://doi.org/10.1016/j.engappai.2025.111412
Received 2 March 2025; Received in revised form 14 May 2025; Accepted 4 June 2025
Available online 20 June 2025
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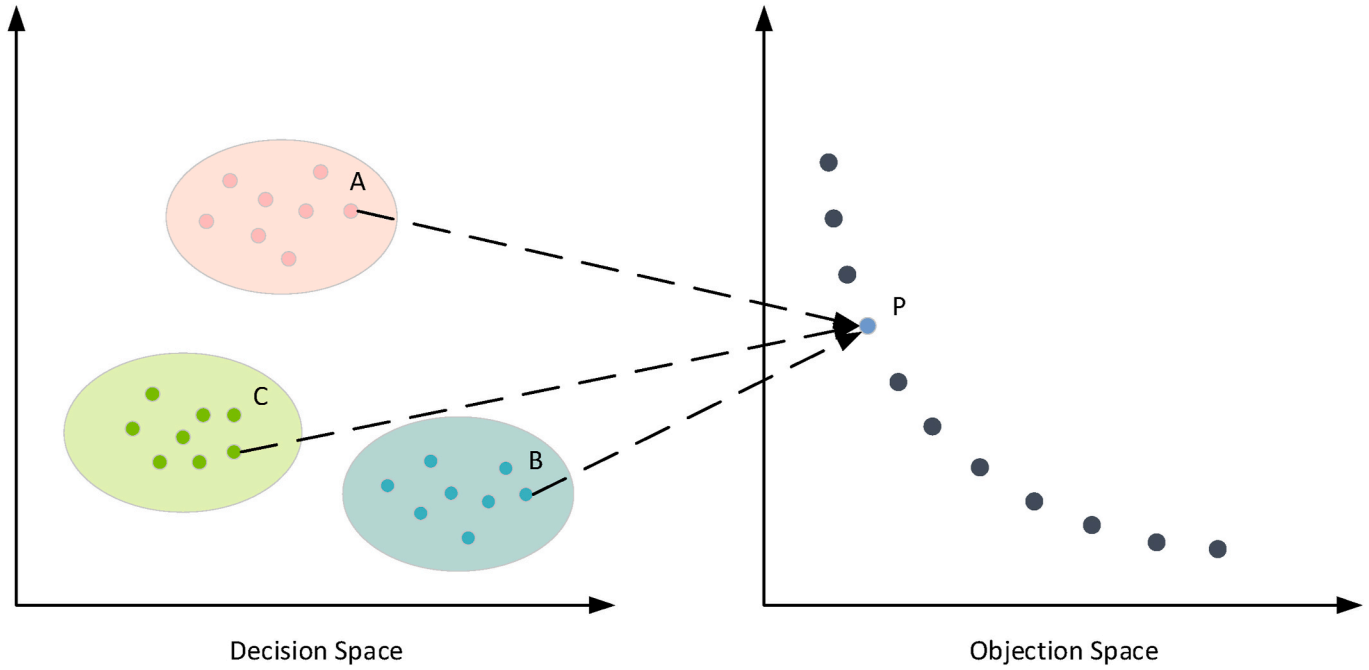


Fig. 1. Schematic diagram of a multimodal multi-objective optimization problem.

to approximate the Pareto optimal set of a given problem through intelligent search techniques. These approaches have been widely applied in practical engineering domains, including fuel cells (Jangir et al., 2024c, 2025), concrete materials (Yue et al., 2024; Mohamed et al., 2024), and food safety (Srikanth et al., 2025). However, such evolutionary algorithms often struggle to effectively address complex multi-objective optimization problems. To address MOPs, many multi-objective evolutionary algorithms (MOEAs) have been proposed (Zhang and Li, 2007; Zitzler et al., 2001; Pandya et al., 2024a, 2024b; Deb et al., 2002; Kalita et al., 2024f, 2024g, 2024h; Ganesh et al., 2023) and have demonstrated effectiveness in practical applications. Most of these algorithms aim to achieve a uniform distribution of the PF within the objective space of MOPs, enabling decision makers (DMs) to select final solutions from the PSs based on personal preferences. Generally, convergence quality is the primary focus of most MOEAs. However, the lack of an effective diversity maintenance mechanism often leads to premature convergence, significantly reducing their search capabilities and limiting their ability to identify superior solutions in complex problems.

As MOPs grow more complex, multimodal multi-objective optimization problems (MMOPs) have become a focal point of research in evolutionary computation. At the same time, they present greater challenges for advanced MOEAs. MMOPs can have multiple distinct PSs corresponding to the same PF. This situation is commonly encountered in the optimization of drug structures (Sakhovskiy and Tutubalina, 2022), feature selection (Yue et al., 2019), microgrid design (Zhang et al., 2023; Li et al., 2019), multi-objective traveler problems (Liu et al., 2023), and other practical applications. In such cases, obtaining the most diverse set of optimal solutions becomes crucial. Taking the multi-objective traveler problem as an example, for the same set of cities, in addition to optimizing the shortest path, multiple objectives such as cost and travel time may also be considered. There are multiple potential combinations of optimal paths, and effectively exploring these different paths while maintaining solution diversity is a core research challenge in this field.

To better understand the multimodal multi-objective optimization problem, a schematic diagram is provided in Fig. 1. As shown in Fig. 1, the decision space is on the left, the objective space is on the right, and there are three points A, B, and C are located in the decision space, each corresponding to point P on the Pareto front. The diagram illustrates that in the decision space, multiple Pareto optimal solution sets can correspond to the same Pareto front. In other words, a solution in the objective space that lies on the Pareto front corresponds to multiple Pareto optimal solutions in the decision space.

MMOPs represent a more challenging research area within multi-objective optimization. In multimodal problems, effectively identifying and maintaining the diversity of multiple solutions is a significant difficulty. Many optimization algorithms tend to focus excessively on a particular objective or region, resulting in a loss of solution diversity.

To address these challenges, several multimodal multi-objective evolutionary algorithms have been proposed. Li et al. designed a local convergence metric to prioritize promising solutions in the search space. By enhancing the environment selection strategy based on population crowding, they improved the population diversity, resulting in the ILC-MMEA (Wen-Hua et al., 2023) algorithm. However, the algorithm tends to emphasize local convergence too strongly, sometimes at the expense of global search capability. Li et al. proposed the algorithm CoMMEA (Li et al., 2023b), which effectively balances convergence and diversity through a coevolutionary strategy, thus maintaining the diversity of the decision space and the convergence of the objective space. Also, the algorithm employs a local convergence metric to balance the distribution of solutions in the objective space. Despite its effectiveness, CoMMEA introduces additional complexity and overhead in managing dual archives and tuning related parameters. Liang et al. proposed the algorithm CMMODE (Liang et al., 2022) which employs a speciation mechanism, in which the first ranked individual in the population is first selected as the seed of the species by a distance ordering, and the nearest k-1 individuals are selected around it to form a complete population of the species and thus to constructing the niche structure in the decision space. Although CMMODE enhances the diversity and feasibility of

feasible Pareto-optimal solutions, it is specifically tailored for constrained multi-objective optimization problems. Zhang et al. proposed a novel bootstrap updating mechanism (Zhang et al., 2019), which incorporates a new decision variable clustering method to divide the population into multiple sub-populations and make them evolve independently to obtain multiple Pareto optimal solutions. To facilitate inter-population interactions, the mechanism employs a ring topology to construct the neighborhoods of the sub-populations. In this framework, evolution within the sub-populations is carried out through global particle swarm optimization, while communication between sub-populations is achieved through local particle swarm optimization. However, it lacks fine-grained guidance in balancing exploration and exploitation across niches, which can hinder convergence speed and accuracy.

Although these algorithms attempt to strike a balance between convergence and diversity through various updating mechanisms, they still face challenges in achieving sufficient convergence to the global optimal solution, maintaining an even population distribution, and ensuring effective global search capability in multimodal multi-objective problems. To address these issues, this paper proposes a multimodal multi-objective optimization algorithm, GOM-MMOEA, which employs a global dynamic orchestration mechanism. The algorithm is innovative in the following aspects:

(1) To improve the convergence of the population, this paper constructs an orchestration mechanism. An orchestration vector is defined, with each element representing the quality of a solution. During each iteration, the algorithm dynamically updates the orchestration vector based on the quality of the solutions. This mechanism effectively guides the search process toward better solutions, thereby accelerating the convergence of the population.
(2) To enhance population diversity and avoid premature convergence, this paper designs a historical orchestration decay strategy for updating the orchestration vector. This strategy gradually diminishes the influence of inferior solutions over time, preventing the population from stagnating in local optima. Moreover, it dynamically adjusts the orchestration factor such that high-quality solutions exert greater influence in early stages to accelerate convergence, while their influence is reduced in later stages to encourage exploration. This balance effectively enhances the global search capability of the algorithm.
(3) To strengthen the exploration ability and maintain population diversity, this paper proposes a novel parent selection mechanism. The mechanism integrates individual orchestration factors with crowding distance and dynamically adjusts the selection probability based on both the current population distribution and historical search information. This dynamic adjustment strategy guides the search toward diverse and promising regions, thereby enhancing the algorithm's global search capability.
(4) To simultaneously promote solution diversity and population convergence, this paper introduces a triple-population synergistic orchestration framework. In this framework, the first population utilizes crowding distance in the objective space to prevent the over-concentration of solutions, the second population employs crowding distance in the decision space to ensure a well-spread distribution of solutions, and the third population combines both objective and decision space distances to achieve more comprehensive global exploration. This synergistic approach effectively maintains a balance between convergence and diversity throughout the entire evolutionary process.

This paper is organized as follows: section I introduces the concepts related to MMOPs; section II introduces the background related to MMOPs and shows the rationality of the algorithms; section III introduces the multimodal multi-objective evolutionary algorithm, GOM-MMOEA, which is proposed in this paper; section IV validates the effectiveness of the proposed algorithm through experiments; and section V concludes the paper.

2. Related work

2.1. Research background

MMOPs have gained significant attention in the field of optimization in recent years. Their unique challenges and wide-ranging applications have made them a focal point of research in both academia and industry. MMOPs not only involve the simultaneous optimization of multiple objective functions but also require the implementation of global exploration during the optimization process to ensure a balanced distribution and diversity of solutions, while effectively balancing the convergence of the objective space with the diversity of the decision space. These complexities make the solution process more intricate, and traditional multi-objective optimization methods (Pandya et al., 2024c; Agrawal et al., 2023) are often inadequate. Therefore, it is of great theoretical value and practical significance to propose an effective optimization algorithm for MMOPs.

An important aspect of MMOPs is the diversity of solutions. Most multimodal multi-objective evolutionary algorithms (MMOEAs) can be broadly categorized into three main groups: decomposition-based strategies (Ma et al., 2017; Wu et al., 2018a, 2018b; Lin et al., 2017; Wang et al., 2016; Jiang et al., 2017; Xu et al., 2017; He et al., 2018) Pareto-based dominance relation strategies (Deb and Jain, 2013; Tian et al., 2018), and indicator-based strategies (Bader and Zitzler, 2011; Tian et al., 2017a; Pamulapati et al., 2018) However, these strategies tend to focus more on assessing the convergence of solutions in the objective space. Unlike traditional multi-objective optimization, MMOPs feature multiple equivalent solutions in the solution space, which can address the diverse needs of DMs. For instance, in building layout design (Michalek et al., 2002), different design solutions may exhibit different advantages and disadvantages in terms of structural strength, cost, material usage, etc., and designers may wish to select among different solutions rather than just obtaining a single globally optimal solution. Therefore, effectively maintaining solution diversity throughout the optimization process, and enabling the algorithm to explore multiple equivalent solutions, has become a key research focus in MMOPs.

The field of multimodal multi-objective optimization can be traced back to early adaptations of traditional multi-objective optimization algorithms. NPGA(Horn et al., 1994), proposed by Horn et al., in 1994, is one of the first algorithms designed to address multi-objective optimization while focusing on maintaining solution diversity. NPGA introduces a subdivision mechanism to identify multiple Pareto-optimal solutions by preserving diversity within the population. However, the algorithm often encounters the issue of insufficient diversity in the generated solutions, which limits its ability to effectively explore the entire objective space.

As research progressed, several algorithms were developed to address the shortcomings of earlier methods. Multimodal Genetic

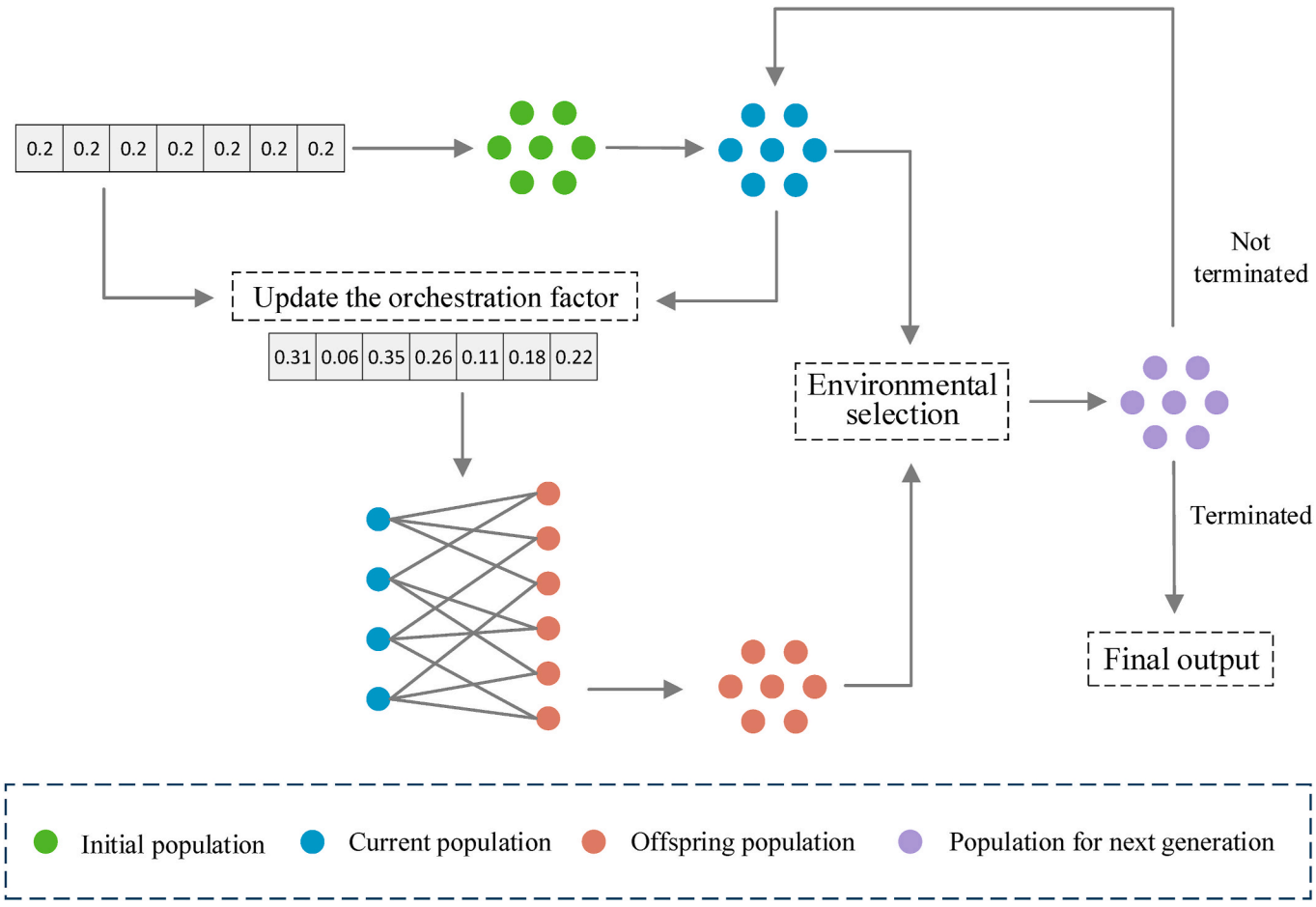


Fig. 2. Flowchart of the GOM-MMOEA algorithm.

Algorithm (MOGA) (Murata and Ishibuchi, 1995) and Multi-Objective Evolutionary Algorithm (MOEA) were introduced to improve the handling of multimodal features. These algorithms typically use hierarchical selection and clustering mechanisms to maintain solution diversity. However, they remain ineffective in generating high-quality solutions, particularly when dealing with the objective space. Specifically, challenges persist in selecting and generating new candidate solutions, making it difficult to identify all the Pareto-optimal solution sets.

Recently, Lin et al. proposed a multimodal multi-objective evolutionary algorithm (MMOEA/DC) (Lin et al., 2020), which uses dual clustering in both the decision space and the objective space. Local Pareto sets are first identified and maintained, followed by the preservation of solution diversity in the objective space. To ensure that only one solution exists per cluster, the most crowded solutions are removed, effectively distinguishing and maintaining both global and local Pareto sets. This approach is particularly well-suited for problems where local Pareto sets coexist with global Pareto sets, as it can effectively differentiate between global Pareto sets and high-quality local Pareto sets.

Liu et al. proposed a novel multimodal multi-objective evolutionary algorithm, TriMOEA-TA&R (Liu et al., 2018), which combines dual-file and recombination strategies to guide the search process by analyzing decision variables and their relationships. The advantage of this algorithm lies in its ability to achieve good convergence and diversity in both the objective and decision spaces, offering an effective framework for solving complex MMOPs.

Li et al. proposed a multimodal multi-objective evolutionary algorithm (MMEA-WI) based on weighted metrics (Li et al., 2021b), which maintains the diversity in the decision space by integrating the information about the diversity of the solutions in the decision space into the objective space performance metrics and introduces a convergence archive to approximate the Pareto-optimal front more efficiently. It better maintains the diversity in the decision space while improving the convergence performance in the objective space. The process of selecting parents is made more efficient by introducing weighted metrics to assess the potential convergence ability of individuals. Li et al. went on to propose an evolutionary algorithm (HREA) (Li et al., 2022) with a hierarchical ordering method, specifically for MMOPs in the presence of a local Pareto front. A hierarchical ordering method is introduced to update the convergence archive, while a local convergence quality assessment method is proposed to better maintain the diversity within the decision space. HREA can find both global and local Pareto fronts based on the DM preferences.

Yue et al. proposed the MO_Ring_PSO_SCD algorithm, which employs a ring topology and a special crowding distance (Yue et al., 2017). The ring topology creates stable niches, allowing for the localization of more Pareto optimal solutions. Additionally, the special crowding distance considers both the crowding distances in the decision space and the objective space, helping to maintain multiple Pareto solution sets. By utilizing the ring topology to form stable niches, the algorithm can identify more Pareto optimal solutions while maintaining the diversity of these solutions through the special crowding distance.

However, as the number of decision variables and the dimensionality of the objective increase, currently available algorithms often struggle to achieve an effective balance between convergence and diversity when addressing MMOPs. Many algorithms are prone to the problem of

premature convergence on MMOPs, which manifests itself as a rapid convergence of the algorithm to a certain localized region in the early stages of the search, thus limiting the exploration of a wider solution space. The root cause of this phenomenon is the insufficient diversity preservation mechanism of existing algorithms, which leads to the inability to discover multiple equivalent solutions efficiently. To avoid such premature convergence, many algorithms try to enhance the breadth of search and exploration by enhancing the diversity in the decision space. Therefore, in multimodal multi-objective optimization, ensuring convergence in the objective space while maintaining diversity in the solution set has become a central challenge in current research.

2.2. Research motivation

Based on the above analysis, it is clear that balancing diversity in the decision space with convergence in the objective space has become a central issue in the current research on MMOEAs.

Therefore, to improve convergence in the objective space while maintaining better diversity in the decision space, ensuring high-quality solution sets, this paper proposes a multimodal multi-objective evolutionary algorithm based on a global orchestration mechanism (GOM-MMOEA). Specifically, the algorithm sets up a vector of orchestration and assigns an orchestration factor to each individual. The orchestration factor optimizes the search process by evaluating the strengths and weaknesses of an individual and guiding its evolutionary direction. The orchestration vector can dynamically guide the distribution of solutions and simultaneously enhance exploration and convergence.

During the search process, the orchestration vector can be dynamically adjusted based on the performance of the solutions, thereby flexibly guiding the search direction and enhancing the algorithm's adaptability in the objective space. To avoid falling into local optimal solutions and enhance the search efficiency, the update of the orchestration vector gradually reduces the strength of the orchestration factor of the inferior solutions through historical orchestration reduction, while strengthening the strength of the orchestration factor of the superior solutions. The increment of the orchestration vector is described in Equation (4) in the next section. In the early stage of evolution, when the number of iterations is small, high-quality solutions dominate a larger number of individuals, resulting in a larger increment of the orchestration factor for these solutions, which effectively accelerates population convergence. In contrast, during the later stages of evolution, as the number of iterations increases, the increment of the orchestration factor for high-quality solutions is gradually reduced, enhancing the global exploration ability and preventing premature convergence. At the same time, the diversity of the population is maintained. This mechanism dynamically adjusts the orchestration vector to balance convergence and diversity, thus improving the global optimization capability of the algorithm.

In addition, the algorithm employs a novel parent selection mechanism that evaluates individual density and orchestration factors to dynamically adjust selection probabilities. This approach flexibly adapts to population distribution, reduces redundant exploration in later stages, and effectively balances convergence and diversity, thereby enhancing search efficiency and global exploration capacity.

The algorithm employs a triple population synergistic orchestration method, specifically, the first population utilizes the density information of the objective space to avoid over-concentration of the solutions in the objective space; the second population ensures uniform distribution of the solutions in the decision space through the density information of the decision space; and the third population further strengthens the global search effect by combining the density information of the objective space and the decision space. Maintaining the diversity of solutions while improving the convergence performance of the populations.

3. Proposed algorithm

3.1. Primary framework

To clearly illustrate the implementation process of the algorithm proposed in this paper, Fig. 2 presents the complete flowchart, providing an intuitive understanding of how the algorithm effectively balances diversity and convergence.

The flowchart shows the core process of the algorithm proposed in this paper. The algorithm generates an initial population in the initial phase and assigns a value to the orchestration factor to each individual. According to the position of an individual in the population, its orchestration factor is updated. During population evolution, individuals are optimized through mating and selection operations, which leads to the environmental selection phase. In this phase, the algorithm screens and updates the population based on fitness assessment to retain high-quality solutions and improve the overall optimization. This process continuously optimizes the population through multiple iterations and promotes a balance between global exploration and convergence.

The overall framework of GOM-MMOEA is presented in Algorithm 1.

Table 1
GOM-MMOEA IGD statistics for representative MOPs at different parameter τ settings.

Problem	M	$\tau = 0.1$	$\tau = 0.3$	$\tau = 0.4$	$\tau = 0.5$	$\tau = 0.2$
MMF1	2	3.3339e-3 (1.70e-4) =	3.3238e-3 (1.75e-4) =	3.3134e-3 (2.51e-4) =	3.2753e-3 (1.53e-4) =	**3.2398e-3 (1.68e-4)**
MMF2	2	1.5777e-2 (4.45e-3) -	1.4139e-2 (2.75e-3) =	1.5806e-2 (5.77e-3) -	1.4911e-2 (3.42e-3) -	**1.3331e-2 (1.98e-3)**
MMF3	2	1.2864e-2 (3.23e-3) -	1.3443e-2 (6.12e-3) -	1.2291e-2 (2.59e-3) -	1.3117e-2 (3.13e-3) -	**1.0179e-2 (1.34e-3)**
MMF4	2	3.2900e-3 (1.37e-4) =	**3.2308e-3 (1.26e-4) =**	3.3011e-3 (1.60e-4) =	3.2652e-3 (1.11e-4) =	3.2359e-3 (1.23e-4)
MMF5	2	3.2998e-3 (1.57e-4) -	**3.2238e-3 (1.80e-4) =**	3.2512e-3 (1.57e-4) =	3.2673e-3 (1.85e-4) =	3.2311e-3 (1.51e-4)
MMF6	2	3.2449e-3 (1.57e-4) =	3.3015e-3 (1.66e-4) =	3.2258e-3 (1.77e-4) =	3.2459e-3 (2.04e-4) =	**3.2159e-3 (1.43e-4)**
MMF7	2	3.1772e-3 (1.49e-4) =	3.2032e-3 (1.41e-4) =	3.2458e-3 (1.50e-4) =	**3.1537e-3 (1.86e-4) =**	3.1725e-3 (1.40e-4)
MMF8	2	3.9433e-3 (2.71e-4) =	4.0369e-3 (3.53e-4) -	4.0066e-3 (2.75e-4) -	3.9247e-3 (2.89e-4) =	**3.8171e-3 (2.21e-4)**
MMMOP1_B	3	1.1556e-1 (2.10e-1) =	**5.5649e-2 (2.39e-2) =**	6.1357e-2 (3.07e-2) =	6.6372e-2 (3.82e-2) =	5.8425e-2 (2.63e-2)
MMMOP2_B	3	**5.7336e-2 (3.59e-3) =**	5.9187e-2 (3.28e-3) =	5.7666e-2 (2.65e-3) =	5.8276e-2 (2.88e-3) =	5.8786e-2 (3.27e-3)
MMMOP3_B	3	3.9875e-2 (1.15e-3) =	3.9669e-2 (1.05e-3) =	3.9845e-2 (1.06e-3) =	3.9545e-2 (1.12e-3) =	**3.9398e-2 (1.28e-3)**
MMMOP4_B	3	6.2013e-1 (4.46e-1) -	4.8363e-1 (2.25e-1) =	5.2074e-1 (3.76e-1) =	6.1601e-1 (3.42e-1) -	**3.8467e-1 (1.94e-1)**
MMMOP5_B	3	1.2203e-1 (6.06e-2) -	1.1285e-1 (5.70e-2) -	1.2332e-1 (6.94e-2) -	1.0849e-1 (4.95e-2) -	**6.8967e-2 (2.57e-2)**
MMMOP6_B	3	3.0766e-2 (2.99e-4) =	**3.0734e-2 (3.68e-4) =**	3.0743e-2 (3.72e-4) =	3.0785e-2 (3.36e-4) =	3.0783e-2 (3.41e-4)
HYL1	2	**2.3121e+0 (3.69e-4) =**	2.3121e+0 (2.63e-4) =	2.3122e+0 (4.19e-4) =	2.3123e+0 (5.01e-4) =	2.3121e+0 (2.83e-4)
HYL2	2	7.0160e+0 (1.44e-3) =	7.0158e+0 (1.09e-3) =	7.0162e+0 (1.53e-3) =	7.0159e+0 (1.00e-3) =	**7.0156e+0 (6.17e-4)**
HYL3	3	9.0021e+0 (1.02e-2) =	9.0027e+0 (1.05e-2) =	**9.0017e+0 (1.10e-2) =**	9.0038e+0 (1.18e-2) =	9.0018e+0 (8.94e-3)
HYL4	4	8.7642e+0 (4.63e-4) =	8.7638e+0 (1.96e-4) =	8.7597e+0 (2.63e-4) =	8.7754e+0 (3.59e-4) =	**8.7582e+0(3.86e-4)**
HYL5	2	6.1355e+0 (7.94e-5) =	6.1355e+0 (1.16e-4) =	6.1355e+0 (1.13e-4) =	**6.1355e+0 (1.19e-4) =**	6.1355e+0 (1.55e-4)
+/−/		0/5/14	0/3/16	0/4/15	0/4/15	

Algorithm 1. Main Framework of GOM-MMOEA

Algorithm 1：Main Framework of GOM-MMOEA

Input: N(Population size), G_{max}(Maximum generations), k(control phase parameters), g(Current number of iterations)

Output: Population P

```
g ← 0;
P ← Initialization(N);//Initial population
τ ← one(N, 1)*0.2 // Initialize the orchestration vector to 0.2 by the experiments in Section 4.3.
while g < G_max do
    if g < k * G_max
        Fitness P ← CalFitness(P, τ);
        OffP ← Variation(ParentP);
        [P, Fitness P] ← Environmental selection(P, OffP, N, τ);
    else if
        //Maintaining population diversity based on Algorithm 2
        P1 ← P;
        P2 ← P;
        [D, Fitness P] ← CalFitness(P, τ);
        [D_dec, Fitness P1] ← CalFitness(P1, τ);
        [D_obj, Fitness P2] ← CalFitness(P2, τ);
        [P, P1, P2] ← Population Diversity(Fitness P, Fitness P1, Fitness P2, D_dec, D_obj, τ, N)
    end if
    τ ← Update the orchestration factor by Equation 2;
    g ← g+1;
end while
return P
```

In the initial stage of the algorithm, a diverse set of solutions is generated to support the subsequent search process, starting with the initial population P and the orchestration factor τ. During the main loop, the algorithm adopts different strategies based on whether the current iteration number g is less than k^*G_{max}, with $k = 0.4$. When $g < k^*G_{max}$, the algorithm gradually guides the population towards the true Pareto front to ensure the convergence of the population. And when $g \geq k^*G_{max}$, the strength of attraction of the solution is orchestrated by dynamically updating the orchestration factor to guide the population. The pre-evolutionary stage strengthens the orchestration influence, thus accelerating the population convergence. The later stage weakens the orchestration influence and enhances the global search capability. By the above method, the GOM-MMOEA algorithm achieves the dual optimization of fast convergence and quality distribution of solutions, which enables the search process to obtain a more uniform and diverse set of solutions while ensuring efficient convergence.

In the above framework, this paper defines the orchestration vector and updates it dynamically. The specific update steps are as follows.

(1) Global Orchestration Update: The update of the orchestration vector is a crucial component of the algorithm. It adjusts the strength of the orchestration factor at the right time by evaluating the performance of the current solution, enabling the algorithm to balance exploration and exploitation, thus improving the ability of global search and convergence efficiency. After each iteration, the algorithm updates the orchestration vector according to the following equation:

$$\tau_{g+1}(i) = \tau_g(i) + \Delta\tau \tag{2}$$

where $\tau_{g+1}(i)$ denotes the intensity of the orchestration factor for the i th individual in the $g+1$ th iteration, $\tau_g(i)$ is the intensity of the orchestration factor for the i th individual in the g th iteration, and $\Delta\tau$ represents the reinforcement of the orchestration increment.

(2) Historical orchestration attenuation: The orchestration factor decays over time, controlled by an evaporation rate that adjusts the impact of historical orchestration information on the evolution of the current individual. This natural decay helps prevent the algorithm from prematurely converging to a local optimum, thus preserving its ability to explore new solutions.

$$\tau_g(i) = (1-\rho)\tau_{g-1}(i) \tag{3}$$

where ρ is the volatility of the orchestration factor, $\rho =$ 0.1. The reduction in historical orchestration causes earlier solutions to gradually lose their attractiveness, prompting the algorithm to continuously update its preferences throughout the search process, thereby providing more opportunities for new candidate solutions.

(3) Enhanced orchestration: The dominance number $S(i)$ of an individual measures its position in the current population, and individuals with larger dominance numbers imply that they are in a relatively superior region in the solution space. In the early stage of the algorithm, the dominance number of the superior solutions is larger, and thus they are given larger increments, which accelerates the convergence process of the population. While in the later stages, the increment gradually decreases and the influence of the high-quality solutions subsequently decreases, which helps to strengthen the global search capability and avoid falling into local optimal solutions.

$$\Delta\tau = \frac{S(i)}{g} \tag{4}$$

$$S(i) = |\{j | j \in P \wedge i \succ j\}| \tag{5}$$

where $S(i)$ denotes the dominance number of the i th individual, and g represents the current iteration number. $\succ$ corresponds to the Pareto dominance relation. In each iteration, a decrease in the strength of the orchestration factor means that the algorithm progressively reduces its reliance on historical solutions, thus encouraging the exploration of new, potentially high-quality solutions. This moderate volatility not only increases the diversity of the search but also improves the efficiency of global optimization, allowing the algorithm to be more flexible in dealing with complex optimization problems. The reinforcement of orchestration cleverly combines changes in the number of individual dominations with the number of iterations. First, it ensures that the top-performing solutions receive more increments in the early stages of evolution, thus accelerating convergence. Second, the increment of high-quality solutions is gradually weakened in the later stages of evolution, reducing the influence of high-quality solutions to avoid the population falling into a local optimum and promoting the maintenance of diversity in the population.

In this study, the fitness of each solution is calculated using the following formula, where the fitness is determined by two factors: $\tau(i)$ and $D(i)$.

$$F(i) = \tau(i) + D(i) \tag{6}$$

$$D(i) = \frac{1}{\delta_i^k + 2} \tag{7}$$

To address the issue of identical fitness values arising from the absence of dominance relationships among individuals, a density measure $D(i)$ is introduced. $D(i)$ represents the density of the region where the individual is located. Its calculation follows the parameter settings specified in (Zitzler et al., 2001), where $k = \sqrt{N+N}$, and δ_i^k is the Euclidean distance from individual i to its k th nearest neighbor.

The fitness value is used as the main criterion for selecting parents, and the strength of the orchestration factor and the density information of the decision space are considered comprehensively. In this way, in the evolutionary process, the individuals in the objective space have better convergence, while the distribution of individuals in the decision space is more uniform.

3.2. Population diversity

While population co-evolution mechanisms have been extensively studied in existing literature, most prior approaches predominantly adopt dual-population cooperative strategies. Typically, one population is designed to enhance convergence in the objective space, while the other promotes diverse distribution in the decision space. This two-population framework has proven effective in balancing convergence and diversity, but may struggle to fully exploit the complex interplay between the objective and decision spaces.

To effectively maintain population diversity and enhance global search capability, this paper adopts a multiple population co-evolution mechanism. This mechanism precisely guides parent selection during the evolutionary process by integrating density information from both the decision space and the objective space. The aim is to ensure a uniform distribution of solutions in the decision space, while also ensuring good distribution and convergence in the objective space. This approach enhances the quality and diversity of the overall solution set. The specific diversity algorithm is shown in Algorithm 2:

Algorithm 2. Population Diversity

Algorithm 2: Population Diversity

Input: Fitness P, Fitness P1, Fitness P2, D_{dec}, D_{obj}, τ, N

Output: P, Fitness P, Fitness P1, Fitness P2

1: Parent P ← Probabilistic selection(P, D, τ); // Based on Algorithm 3
1: OffP ← Variation(Parent P);
2: Parent P1 ← Probabilistic selection(P1, D_{dec}, τ);
3: OffP1 ← Variation(Parent P1);
4: Parent P2 ← Probabilistic selection(P2, D_{obj}, τ);
5: OffP2 ← Variation(Parent P2);
6: OffP ← [OffP, OffP1, OffP2];
7: [P, Fitness P, D] ← Environmental selection(P, OffP, N, τ);
8: [P1, Fitness P1, D_{dec}] ← Environmental selection(P1, OffP1, N, τ);
9: [P2, Fitness P2, D_{obj}] ← Environmental selection(P2, OffP2, N, τ);
10: **return** P, P1, P2

Population P integrates the density information from both the decision space and the objective space to guide parent selection, thereby maintaining population diversity during the evolutionary process and ensuring both solution convergence and global search capability. Population $P1$ primarily utilizes density information from the decision space for parent selection, ensuring a uniform distribution of solutions within the decision space. On the other hand, population $P2$ incorporates density information from the objective space to optimize parent selection, ensuring that solutions in the objective space maintain a uniform distribution and achieve good convergence. Through this multi-population co-evolution mechanism, the GOM-MMOEA algorithm achieves balanced optimization in both spaces, preventing uneven distribution in the decision space while ensuring uniform distribution in the objective space. This significantly enhances the algorithm's global search ability and the quality of the solutions.

3.3. Parental selection mechanisms

This paper proposes a new parent selection mechanism that combines the orchestration factor strength with density information. As indicated in Equation (4), the orchestration factor is used to measure an individual's convergence, reflecting its degree of convergence in the objective space, while the density information evaluates the distribution of individuals in the decision space, representing the diversity of the population. This mechanism effectively balances the convergence and diversity of individuals. It can dynamically adjust the parent selection preference based on the distribution of the population.

$$\gamma(i) = \frac{\tau(i)D(i)}{\Sigma_{j=1}^{N}\tau(j)D(j)} \tag{8}$$

where $\gamma(i)$ denotes the selection probability of the i th individual, which is influenced by both the orchestration factor and the density information. This algorithm, firstly, calculates the selection probability of each individual; secondly, the algorithm dynamically adjusts the selection tendency according to the position of the individual as well as the difference of the orchestration factor; finally, the algorithm selects the high-quality solution with a high probability from the parent generation, thus improving the overall search efficiency.

To enhance the diversity of the search, the algorithm forms a mating pool by sampling individuals from the population based on the calculated selection probability. This approach ensures that even individuals with low selection probability have a certain chance to participate in the subsequent crossover and mutation operations, thus effectively avoiding the search process from falling into the dilemma of local optimal solutions.

Algorithm 3. Probabilistic selection

Algorithm 3: Probabilistic selection

Input: P, D, τ

Output: P

1: **for i = 1:|P|**

2: $\gamma(i)$ ← Calculate the probability of an individual being selected by Equation 8;

3: **end**

4: Mating Pool ← sample(1:|P|, |P|, γ)

5: **return** P

3.4. Environmental selection

In the environmental selection phase, the original population is first combined with the offspring to form a joint file Union, and then the fitness values of the individuals are calculated based on this joint file Union (lines 1 and 2). Special attention should be paid to the fact that in the algorithm, when Fitness Union <1, the individual is recognized as a

Table 2
GOM-MMOEA IGDX statistics for representative MOPs at different parameter τ settings.

Problem	M	$\tau = 0.1$	$\tau = 0.3$	$\tau = 0.4$	$\tau = 0.5$	$\tau = 0.2$
MMF1	2	4.7281e-2 (3.51e-3) -	4.6710e-2 (4.30e-3) -	4.7231e-2 (3.42e-3) -	4.7859e-2 (3.90e-3) -	**4.1941e-2 (1.28e-3)**
MMF2	2	4.1093e-2 (1.44e-2) -	3.8367e-2 (1.26e-2) -	4.1212e-2 (1.33e-2) -	4.3992e-2 (1.74e-2) -	**3.1702e-2 (9.55e-3)**
MMF3	2	3.7376e-2 (1.37e-2) -	3.8562e-2 (1.54e-2) -	3.4631e-2 (1.15e-2) -	3.7177e-2 (1.53e-2) -	**2.3907e-2 (2.30e-3)**
MMF4	2	2.2624e-2 (2.57e-3) -	2.2504e-2 (1.94e-3) -	2.2872e-2 (1.97e-3) -	2.2638e-2 (2.30e-3) -	**2.0388e-2 (5.07e-4)**
MMF5	2	7.9767e-2 (6.27e-3) -	7.8174e-2 (5.09e-3) -	8.0462e-2 (7.05e-3) -	7.9276e-2 (6.01e-3) -	**7.1354e-2 (2.16e-3)**
MMF6	2	6.7771e-2 (4.89e-3) -	6.7796e-2 (6.63e-3) -	6.7365e-2 (6.46e-3) -	6.6111e-2 (3.66e-3) -	**6.0277e-2 (8.67e-4)**
MMF7	2	2.3126e-2 (3.43e-3) -	2.4305e-2 (4.05e-3) -	2.3603e-2 (3.87e-3) -	2.3525e-2 (3.97e-3) -	**1.9793e-2 (8.49e-4)**
MMF8	2	1.2840e-1 (5.80e-2) -	1.2801e-1 (4.45e-2) -	1.4230e-1 (4.38e-2) -	1.3029e-1 (4.61e-2) -	**8.7678e-2 (1.52e-2)**
MMMOP1_B	3	1.9028e-1 (4.68e-2) -	1.8201e-1 (4.94e-2) -	2.0814e-1 (5.42e-2) -	1.9414e-1 (4.82e-2) -	**1.4103e-1 (3.02e-2)**
MMMOP2_B	3	1.2757e-1 (9.45e-2) -	1.3139e-1 (9.52e-2) -	1.3516e-1 (9.73e-2) -	1.1022e-1 (5.45e-2) -	**5.5444e-2 (6.74e-3)**
MMMOP3_B	3	8.6081e-2 (2.60e-3) -	8.5494e-2 (1.86e-3) -	8.6538e-2 (2.02e-3) -	8.5586e-2 (2.44e-3) -	**8.3514e-2 (1.47e-3)**
MMMOP4_B	3	1.8422e-1 (2.23e-2) -	1.8254e-1 (2.58e-2) -	1.8429e-1 (2.49e-2) -	1.8964e-1 (2.81e-2) -	**1.5132e-1 (1.50e-2)**
MMMOP5_B	3	1.0940e-1 (3.17e-2) -	1.0565e-1 (3.08e-2) -	1.1765e-1 (3.22e-2) -	1.1076e-1 (2.57e-2) -	**7.8408e-2 (1.10e-2)**
MMMOP6_B	3	3.9358e-2 (5.90e-4) -	3.9348e-2 (4.64e-4) -	3.9476e-2 (5.53e-4) -	3.9447e-2 (6.63e-4) -	**3.8752e-2 (2.49e-4)**
HYL1	2	1.9030e-1 (7.98e-3) =	**1.8804e-1 (9.61e-4)** =	1.8863e-1 (1.52e-3) =	1.9633e-1 (3.97e-2) =	1.8842e-1 (2.11e-3)
HYL2	2	3.8785e-1 (2.28e-3) =	3.8746e-1 (2.01e-3) =	3.8820e-1 (2.03e-3) =	**3.8706e-1 (2.20e-3)** =	3.8733e-1 (2.32e-3)
HYL3	3	6.3217e-1 (5.62e-3) =	**6.3105e-1 (5.42e-3)** =	6.3463e-1 (5.87e-3) =	6.3273e-1 (6.96e-3) =	6.3477e-1 (7.19e-3)
HYL4	4	9.1801e-1 (5.48e-3) =	**9.1699e-1 (7.29e-3)** =	9.1885e-1 (6.68e-3) =	9.1847e-1 (6.56e-3) =	9.1961e-1 (7.86e-3)
HYL5	2	1.6975e-1 (8.09e-4) =	1.6954e-1 (7.56e-4) +	1.6968e-1 (5.60e-4) +	**1.6933e-1 (9.51e-4)** +	1.7013e-1 (8.17e-4)
+/−/ =		0/14/5	1/14/4	1/14/4	1/14/4	

non-dominated solution. If the number of non-dominated solutions is less than the population size N, the solutions are ranked by fitness value and crowding distance, and the top N solutions are selected to form a new population; if the number of non-dominated solutions is more than N, the truncation method (shown in Eq. (9)) based on crowding distance in (Lin et al., 2020) is used to keep the population size constant.

In multimodal multi-objective optimization, various diversity preservation strategies—such as niche preservation, density estimation, and clustering-based techniques—have been proposed. Niche preservation prevents individuals from clustering too closely and promotes solution spread across the objective space, while clustering-based methods group similar individuals to explore distinct solution regions. However, these methods often require predefined parameters or prior knowledge of the problem landscape, which can be difficult to obtain in complex scenarios. To overcome these limitations, our algorithm employs the crowding distance strategy from MMOEA/DC for environmental selection, enabling adaptive density estimation without the need for additional parameters. This method effectively maintains a balance between diversity and convergence, while keeping computational complexity low.

$$CrowdDis_i = \frac{N-1}{\frac{\sum_{j=1}^{N} 1}{\left\|x_j - x_i\right\|}} \tag{9}$$

where $\left\|x_j - x_i\right\|$ denotes the Euclidean distance between solutions x_i and x_j.

Algorithm 4. Environmental Selection Framework

Algorithm 4: Environmental Selection Framework

Input: P, OffP, N, τ
Output: P, Fitness P, D_{dec}, D_{obj}

```
Union ← P ∪ OffP;
[D_dec, D_obj, Fitness Union] ← CalFitness(Union, τ);
Next ← Fitness Union < 1;
if Num (Next) < N
    IDX ← sort(Fitness Union);
    P ← Union(IDX(1:N));
else
    P ← Truncate(Union (Next), N);
end
return P, Fitness P, D_dec, D_obj
```

4. Experimental results and analysis

4.1. Experimental setting

This chapter focuses on evaluating and comparing the performance of GOM-MMOEA against five competing algorithms (MO_Ring_PSO_SCD (Yue et al., 2017), ILC-MMEA (Wen-Hua et al., 2023), MMEA-WI (Li et al., 2021b), MMOEA/DC (Lin et al., 2020), HREA (Li et al., 2022)). To achieve this, three types of test problems are selected: MMF(Yue et al., 2017), MMMOP (Liu et al., 2018), and HYL (Ishibuchi et al., 2022). Among them, the MMMOP test suite includes MMOPs with two to seven decision variables and two to three objective functions. Some of these problems feature multiple equivalent Pareto solution sets, which pose challenges for multimodal multi-objective evolutionary algorithms in exploring the decision space. The HYL test suite, a recently proposed benchmark test set, contains MOPs with an infinite number of equivalent Pareto optimal solutions. These problems are selected to thoroughly evaluate the effectiveness and adaptability of the algorithms in multimodal multi-objective optimization scenarios. During the experiments, to ensure the scientific validity and reliability of the results, each test problem was run independently 30 times. The Wilcoxon rank-sum test was used to assess the significant differences between GOM-MMOEA and the five comparison algorithms. In the following tables, + after the statistics indicates that the comparison algorithms outperform GOM-MMOEA, - indicates that GOM-MMOEA outperforms the comparison algorithms, and = indicates that GOM-MMOEA is comparable to the comparative algorithm. All experiments were performed on PlatEMO(Tian et al., 2017b). The population sizes and maximum number of evaluations in the three test scenario experiments were determined based on information provided by the authors of the original paper, ensuring that the computational resource requirements for each comparison algorithm were set as follows:

- MMF: N = 200 E_{max} = 10000;
- MMMOP: N = 400 E_{max} = 50000;
- HYL: N = 200 E_{max} = 20000;

4.2. Performance indicators

In MMOPs, performance metrics are crucial for assessing the effectiveness of an algorithm. Traditional multi-objective optimization algorithms primarily focus on finding a set of solutions that closely approximates the true Pareto front in the objective space. IGD is an effective metric for measuring the algorithm's convergence, reflecting its ability to approach the optimal solution. However, in MMOPs, considering only the convergence of the solution set in the objective space is insufficient. It is also essential to evaluate the distribution of solutions in the decision space. MMOPs require that the solution set not only approximates the Pareto front but also maintains a uniform distribution in the decision space to capture multiple distinct equivalent solutions. In this context, IGDX, which measures the uniformity of the solution set's distribution in the decision space, serves as an effective metric to assess the algorithm's diversity.

IGD is a widely used performance metric in multi-objective optimization for the closeness between the Pareto front obtained by the algo-

Table 3
GOM-MMOEA IGD statistics for representative MOPs at different parameter ρ settings.

Problem	M	$\rho = 0.2$	$\rho = 0.3$	$\rho = 0.4$	$\rho = 0.5$	$\rho = 0.1$
MMF1	2	3.3038e-3 (1.61e-4) =	3.3028e-3 (1.72e-4) =	3.2779e-3 (1.72e-4) =	3.3409e-3 (1.81e-4) -	**3.2398e-3 (1.68e-4)**
MMF2	2	1.5788e-2 (5.30e-3) =	1.4870e-2 (4.22e-3) =	1.6087e-2 (4.79e-3) -	1.6904e-2 (6.22e-3) -	**1.3331e-2 (1.98e-3)**
MMF3	2	1.2774e-2 (3.57e-3) -	1.4534e-2 (6.62e-3) -	1.2266e-2 (3.06e-3) -	1.2623e-2 (3.27e-3) -	**1.0179e-2 (1.34e-3)**
MMF4	2	3.2635e-3 (1.33e-4) =	3.2788e-3 (1.24e-4) =	3.2590e-3 (1.21e-4) =	3.2545e-3 (1.21e-4) =	**3.2359e-3 (1.23e-4)**
MMF5	2	3.2510e-3 (1.92e-4) =	3.3409e-3 (1.66e-4) -	3.2789e-3 (1.49e-4) =	3.3250e-3 (2.19e-4) =	**3.2311e-3 (1.51e-4)**
MMF6	2	3.2611e-3 (1.48e-4) =	3.2427e-3 (1.48e-4) =	3.2760e-3 (1.89e-4) =	3.2315e-3 (1.79e-4) =	**3.2159e-3 (1.43e-4)**
MMF7	2	3.2115e-3 (1.72e-4) =	**3.1573e-3 (1.48e-4) =**	3.2105e-3 (1.97e-4) =	3.1904e-3 (1.77e-4) =	3.1725e-3 (1.40e-4)
MMF8	2	3.9329e-3 (2.43e-4) =	3.9967e-3 (3.15e-4) =	3.9256e-3 (2.31e-4) =	3.9309e-3 (1.88e-4) =	**3.8171e-3 (2.21e-4)**
MMMOP1_B	3	**5.0825e-2 (2.91e-2) =**	6.0971e-2 (3.17e-2) =	5.4129e-2 (3.13e-2) =	5.7693e-2 (2.98e-2) =	5.8425e-2 (2.63e-2)
MMMOP2_B	3	5.9574e-2 (2.98e-3) =	5.8780e-2 (2.88e-3) =	**5.8427e-2 (2.63e-3) =**	5.9346e-2 (3.14e-3) =	5.8786e-2 (3.27e-3)
MMMOP3_B	3	3.9667e-2 (1.02e-3) =	**3.9364e-2 (1.01e-3) =**	3.9744e-2 (1.03e-3) =	3.9874e-2 (7.82e-4) -	3.9398e-2 (1.28e-3)
MMMOP4_B	3	5.4813e-1 (3.37e-1) -	5.3946e-1 (3.51e-1) =	6.3082e-1 (4.24e-1) -	5.7317e-1 (3.94e-1) -	**3.8467e-1 (1.94e-1)**
MMMOP5_B	3	1.5050e-1 (1.36e-1) -	1.5819e-1 (1.73e-1) -	1.3414e-1 (8.82e-2) -	1.4815e-1 (1.57e-1) -	**6.8967e-2 (2.57e-2)**
MMMOP6_B	3	**3.0633e-2 (3.48e-4) =**	3.0689e-2 (3.24e-4) =	3.0785e-2 (3.68e-4) =	3.0751e-2 (3.68e-4) =	3.0783e-2 (3.41e-4)
HYL1	2	2.3124e+0 (6.53e-4) =	2.3122e+0 (3.76e-4) =	2.3122e+0 (3.95e-4) =	2.3123e+0 (5.67e-4) =	**2.3121e+0 (2.83e-4)**
HYL2	2	7.0157e+0 (7.39e-4) =	7.0160e+0 (1.04e-3) =	7.0158e+0 (7.81e-4) =	7.0160e+0 (1.15e-3) =	**7.0156e+0 (6.17e-4)**
HYL3	3	**8.9994e+0 (8.67e-3) =**	9.0021e+0 (1.07e-2) =	9.0009e+0 (1.03e-2) =	9.0015e+0 (1.20e-2) =	9.0018e+0 (8.94e-3)
HYL4	4	8.7593e+0 (3.84e-3) =	8.7589e+0 (2.47e-3) =	8.7652e+0 (5.92e-3) =	8.7603e+0 (1.35e-3) =	**8.7582e+0(3.86e-4)**
HYL5	2	**6.1355e+0 (8.89e-5) =**	6.1355e+0 (1.34e-4) =	6.1355e+0 (1.26e-4) =	6.1356e+0 (1.96e-4) =	6.1355e+0 (1.55e-4)
+/−/ =		0/3/16	0/3/16	0/4/15	0/6/13	

rithm and the true Pareto front (Coello Coello and Reyes Sierra, 2004). It is formulated as follows:

$$IGD(S, P^*) = \frac{1}{|P^*|} \sum_{p^* \in P^*} \min_{s \in S} d(p^*, s) \tag{10}$$

where P^* denotes the set of reference points of the Pareto front, S denotes the set of solutions found by the algorithm, and $d(p^*, s)$ is the distance between the solution s and the reference point p^*, which is usually measured using the Euclidean Distance metric. The smaller the value of the IGD, the closer the distance between the set of solutions and the true Pareto front is, and the better the algorithm performs.

IGDX is used to measure the closeness between the solution set and the true Pareto solution set (Zhou et al., 2009) with the following formula:

$$IGDX(S, P_D^*) = \frac{1}{|P_D^*|} \sum_{p_D^* \in P_D^*} \min_{s \in S} d(p_D^*, s) \tag{11}$$

where P_D^* denotes the equivalent solution set in the decision space, and $d(p_D^*, s)$ denotes the distance between the solution s and the equivalent solution p_D^*. The smaller the value of IGDX, the closer the solution set obtained by the algorithm is to the true Pareto solution set.

4.3. Parametric analysis of the GOM-MMOEA

In this section, the key parameters in the proposed GOM-MMOEA algorithm are comprehensively investigated, focusing on the roles of the initialization orchestration factor τ and the decay rate ρ of the orchestration factor. To assess the impact of these parameters on the performance of the algorithm, experiments are conducted and analyzed for representative problems with different parameter settings. Regarding the MMMOP suite, we specifically presented results for the _B variants (e.g., MMMOP1-B, MMMOP2-B, etc.) rather than all variants. The MMMOP-B problems were chosen as representative cases because they exhibit higher dimensionality in both objectives and decision variables, increasing the complexity of the search space and posing greater challenges for maintaining diversity and convergence simultaneously.

1) Analysis of the initialization orchestration factor τ

The initialization τ plays a crucial role in the algorithm, which assigns an initial value to each individual and is used to orchestrate the direction of individual evolution. When the initial value of τ is set too

Table 4
GOM-MMOEA IGDX statistics for representative MOPs at different parameter ρ settings.

Problem	M	$\rho = 0.2$	$\rho = 0.3$	$\rho = 0.4$	$\rho = 0.5$	$\rho = 0.1$
MMF1	2	4.7080e-2 (5.16e-3) -	4.7593e-2 (3.41e-3) -	4.6787e-2 (5.15e-3) -	4.5401e-2 (3.11e-3) -	**4.1941e-2 (1.28e-3)**
MMF2	2	4.2713e-2 (1.72e-2) -	4.7934e-2 (3.09e-2) -	4.7431e-2 (1.54e-2) -	4.9760e-2 (1.89e-2) -	**3.1702e-2 (9.55e-3)**
MMF3	2	3.4929e-2 (1.22e-2) -	3.7798e-2 (1.02e-2) -	3.2637e-2 (9.74e-3) -	3.5081e-2 (1.13e-2) -	**2.3907e-2 (2.30e-3)**
MMF4	2	2.3444e-2 (2.58e-3) -	2.2944e-2 (2.54e-3) -	2.3067e-2 (1.83e-3) -	2.4536e-2 (3.53e-3) -	**2.0388e-2 (5.07e-4)**
MMF5	2	7.9700e-2 (6.19e-3) -	7.8803e-2 (7.71e-3) -	8.0763e-2 (7.49e-3) -	8.0200e-2 (6.95e-3) -	**7.1354e-2 (2.16e-3)**
MMF6	2	6.8050e-2 (6.03e-3) -	6.6403e-2 (5.36e-3) -	6.6972e-2 (5.58e-3) -	6.7785e-2 (5.96e-3) -	**6.0277e-2 (8.67e-4)**
MMF7	2	2.3164e-2 (3.69e-3) -	2.3363e-2 (3.25e-3) -	2.3286e-2 (3.17e-3) -	2.2405e-2 (2.72e-3) -	**1.9793e-2 (8.49e-4)**
MMF8	2	1.2689e-1 (4.96e-2) -	1.4126e-1 (5.49e-2) -	1.3853e-1 (4.04e-2) -	1.3965e-1 (4.68e-2) -	**8.7678e-2 (1.52e-2)**
MMMOP1_B	3	1.9869e-1 (6.30e-2) -	1.9038e-1 (4.75e-2) -	1.8910e-1 (6.95e-2) -	2.1222e-1 (8.04e-2) -	**1.4103e-1 (3.02e-2)**
MMMOP2_B	3	1.0001e-1 (5.94e-2) -	1.2578e-1 (1.00e-1) -	1.3089e-1 (9.70e-2) -	1.2248e-1 (6.97e-2) -	**5.5444e-2 (6.74e-3)**
MMMOP3_B	3	8.5964e-2 (1.93e-3) -	8.5876e-2 (2.20e-3) -	8.5824e-2 (2.19e-3) -	8.6006e-2 (2.23e-3) -	**8.3514e-2 (1.47e-3)**
MMMOP4_B	3	1.8341e-1 (3.12e-2) -	1.8167e-1 (2.62e-2) -	1.7675e-1 (3.04e-2) -	1.9242e-1 (1.90e-2) -	**1.5132e-1 (1.50e-2)**
MMMOP5_B	3	1.2024e-1 (2.83e-2) -	1.1432e-1 (2.87e-2) -	1.1404e-1 (3.09e-2) -	1.1443e-1 (3.81e-2) -	**7.8408e-2 (1.10e-2)**
MMMOP6_B	3	3.9570e-2 (8.41e-4) -	3.9219e-2 (5.16e-4) -	3.9559e-2 (7.83e-4) -	3.9465e-2 (5.28e-4) -	**3.8752e-2 (2.49e-4)**
HYL1	2	**1.8821e-1 (2.00e-3) =**	1.9160e-1 (1.57e-2) =	1.8841e-1 (1.42e-3) =	1.9222e-1 (2.18e-2) =	1.8842e-1 (2.11e-3)
HYL2	2	3.8757e-1 (1.87e-3) =	**3.8686e-1 (2.18e-3) =**	3.8831e-1 (1.90e-3) =	3.8766e-1 (1.90e-3) =	3.8733e-1 (2.32e-3)
HYL3	3	**6.3354e-1 (5.85e-3) =**	6.3424e-1 (6.08e-3) =	6.3386e-1 (5.72e-3) =	6.3398e-1 (5.50e-3) =	6.3477e-1 (7.19e-3)
HYL4	4	**9.1775e-1 (8.03e-3) =**	9.1801e-1 (9.08e-3) =	9.1791e-1 (7.46e-3) =	9.2022e-1 (7.42e-3) =	9.1961e-1 (7.86e-3)
HYL5	2	1.6978e-1 (1.02e-3) =	1.6942e-1 (6.38e-4) +	1.6953e-1 (8.24e-4) +	**1.6934e-1 (7.17e-4) +**	1.7013e-1 (8.17e-4)
+/−/ =		0/14/5	1/14/4	1/14/4	1/14/4	

Table 5
GOM-MMOEA IGD statistics for representative MOPs at different parameter k settings.

Problem	M	k = 0.1	k = 0.2	k = 0.3	k = 0.5	k = 0.4
MMF1	2	3.3503e-3 (2.11e-4) =	3.3695e-3 (1.96e-4) -	3.3086e-3 (2.03e-4) =	3.2655e-3 (2.21e-4) =	**3.2398e-3 (1.68e-4)**
MMF2	2	1.3864e-2 (2.59e-3) =	1.4104e-2 (5.31e-3) =	1.4737e-2 (4.88e-3) =	1.5543e-2 (4.10e-3) -	**1.3331e-2 (1.98e-3)**
MMF3	2	1.2078e-2 (2.28e-3) -	1.1989e-2 (1.69e-3) -	1.1677e-2 (2.03e-3) -	1.3163e-2 (3.44e-3) -	**1.0179e-2 (1.34e-3)**
MMF4	2	3.2543e-3 (1.34e-4) =	3.3316e-3 (1.75e-4) -	3.2865e-3 (1.50e-4) =	3.3127e-3 (1.41e-4) =	**3.2359e-3 (1.23e-4)**
MMF5	2	3.3401e-3 (1.59e-4) -	3.3279e-3 (2.15e-4) =	3.3376e-3 (2.58e-4) =	3.2907e-3 (2.10e-4) =	**3.2311e-3 (1.51e-4)**
MMF6	2	3.3393e-3 (1.41e-4) -	3.2886e-3 (2.26e-4) =	3.2610e-3 (1.47e-4) =	**3.1808e-3 (1.29e-4) =**	3.2159e-3 (1.43e-4)
MMF7	2	3.2545e-3 (1.32e-4) -	3.2618e-3 (1.96e-4) =	3.2569e-3 (1.74e-4) =	3.1742e-3 (1.72e-4) =	**3.1725e-3 (1.40e-4)**
MMF8	2	4.0194e-3 (2.35e-4) -	4.1674e-3 (3.23e-4) -	3.9950e-3 (2.69e-4) -	3.9400e-3 (3.42e-4) =	**3.8171e-3 (2.21e-4)**
MMMOP1_B	3	4.0409e-1 (2.29e-1) -	1.9474e-1 (1.01e-1) -	1.3402e-1 (1.14e-1) -	**3.9306e-2 (1.61e-2) +**	5.8425e-2 (2.63e-2)
MMMOP2_B	3	**5.7778e-2 (2.64e-3) =**	8.5132e-2 (1.49e-1) =	5.8159e-2 (2.75e-3) =	5.8026e-2 (1.87e-3) =	5.8786e-2 (3.27e-3)
MMMOP3_B	3	3.9424e-2 (1.02e-3) =	3.9898e-2 (8.91e-4) -	3.9439e-2 (1.35e-3) =	3.9653e-2 (1.16e-3) =	**3.9398e-2 (1.28e-3)**
MMMOP4_B	3	1.9021e+0 (8.03e-1) -	1.2904e+0 (6.27e-1) -	9.8790e-1 (5.67e-1) -	3.9191e-1 (2.28e-1) =	**3.8467e-1 (1.94e-1)**
MMMOP5_B	3	1.0578e+0 (1.25e+0) -	5.1769e-1 (3.78e-1) -	1.9769e-1 (9.98e-2) -	**5.5479e-2 (2.37e-2) +**	6.8967e-2 (2.57e-2)
MMMOP6_B	3	3.0982e-2 (3.65e-4) -	3.0847e-2 (3.21e-4) =	3.0770e-2 (3.60e-4) =	**3.0664e-2 (3.31e-4) =**	3.0783e-2 (3.41e-4)
HYL1	2	2.3121e+0 (2.79e-4) =	2.3123e+0 (4.56e-4) =	2.3122e+0 (4.45e-4) =	2.3121e+0 (5.72e-4) =	**2.3121e+0 (2.83e-4)**
HYL2	2	7.0158e+0 (9.21e-4) =	7.0158e+0 (8.49e-4) =	7.0159e+0 (9.74e-4) =	2.3121e+0 (5.72e-4) =	**7.0156e+0 (6.17e-4)**
HYL3	3	9.0016e+0 (9.95e-3) =	**8.9991e+0 (6.53e-3) =**	9.0025e+0 (1.00e-2) =	7.0164e+0 (1.71e-3) -	9.0018e+0 (8.94e-3)
HYL4	4	8.7598e+0 (4.42e-3) =	8.7677e+0 (3.42e-3) =	8.7593e+0 (4.78e-3) =	9.0026e+0 (1.12e-2) =	**8.7582e+0(3.86e-4)**
HYL5	2	6.1355e+0 (1.01e-4) =	6.1356e+0 (1.13e-4) =	6.1355e+0 (1.10e-4) =	8.7664e+0 (4.65e-3) =	**6.1355e+0 (1.55e-4)**
+/−/ =		0/9/10	0/8/11	0/5/14	2/3/14	

Table 6
GOM-MMOEA IGDX statistics for representative MOPs at different parameter k settings.

IGDX/k	M	k = 0.1	k = 0.2	k = 0.3	k = 0.5	k = 0.4
MMF1	2	**4.1395e-2 (3.02e-3) =**	4.3078e-2 (2.78e-3) -	4.6488e-2 (5.61e-3) -	4.6702e-2 (4.18e-3) -	4.1941e-2 (1.28e-3)
MMF2	2	**3.0514e-2 (6.02e-3) =**	3.2494e-2 (8.35e-3) =	3.8874e-2 (1.53e-2) -	4.6141e-2 (2.10e-2) -	3.1702e-2 (9.55e-3)
MMF3	2	2.7194e-2 (4.54e-3) -	3.2236e-2 (1.19e-2) -	3.3695e-2 (8.76e-3) -	3.6667e-2 (8.46e-3) -	**2.3907e-2 (2.30e-3)**
MMF4	2	2.0765e-2 (1.57e-3) =	2.1300e-2 (1.97e-3) =	2.1852e-2 (1.70e-3) -	2.3962e-2 (2.71e-3) -	**2.0388e-2 (5.07e-4)**
MMF5	2	7.1466e-2 (4.03e-3) =	7.4041e-2 (5.46e-3) =	7.6877e-2 (8.67e-3) -	7.9797e-2 (6.67e-3) -	**7.1354e-2 (2.16e-3)**
MMF6	2	6.0684e-2 (3.09e-3) =	6.2313e-2 (5.13e-3) =	6.5300e-2 (4.93e-3) -	6.9879e-2 (6.40e-3) -	**6.0277e-2 (8.67e-4)**
MMF7	2	2.0884e-2 (2.10e-3) =	2.1834e-2 (3.03e-3) -	2.3197e-2 (2.98e-3) -	2.4093e-2 (3.85e-3) -	**1.9793e-2 (8.49e-4)**
MMF8	2	**7.4267e-2 (2.14e-2) +**	8.9619e-2 (2.14e-2) =	1.0964e-1 (3.65e-2) -	1.5294e-1 (6.21e-2) -	8.7678e-2 (1.52e-2)
MMMOP1_B	3	1.8822e-1 (3.92e-2) -	1.9247e-1 (4.47e-2) -	1.8649e-1 (5.14e-2) -	1.7290e-1 (6.10e-2) -	**1.4103e-1 (3.02e-2)**
MMMOP2_B	3	6.1403e-2 (1.83e-2) =	9.2524e-2 (8.04e-2) -	1.0142e-1 (6.55e-2) -	1.4012e-1 (6.59e-2) -	**5.5444e-2 (6.74e-3)**
MMMOP3_B	3	8.8580e-2 (2.33e-3) -	8.6925e-2 (1.86e-3) -	8.6085e-2 (2.00e-3) -	8.6059e-2 (1.81e-3) -	**8.3514e-2 (1.47e-3)**
MMMOP4_B	3	2.0202e-1 (1.95e-2) -	1.8565e-1 (2.64e-2) -	1.8402e-1 (2.30e-2) -	**1.4378e-1 (2.62e-2) =**	1.5132e-1 (1.50e-2)
MMMOP5_B	3	1.7677e-1 (3.14e-2) -	1.6302e-1 (3.15e-2) -	1.4912e-1 (3.27e-2) -	8.2645e-2 (2.27e-2) =	**7.8408e-2 (1.10e-2)**
MMMOP6_B	3	**3.8361e-2 (4.06e-4) +**	3.8798e-2 (5.37e-4) =	3.9049e-2 (5.38e-4) -	3.9688e-2 (6.76e-4) -	3.8752e-2 (2.49e-4)
HYL1	2	**1.8796e-1 (9.22e-4) =**	1.8851e-1 (1.67e-3) =	1.8835e-1 (1.34e-3) =	1.8923e-1 (5.46e-3) =	1.8842e-1 (2.11e-3)
HYL2	2	3.8765e-1 (1.88e-3) =	3.8804e-1 (2.56e-3) =	3.8784e-1 (2.28e-3) =	3.8743e-1 (2.22e-3) =	**3.8733e-1 (2.32e-3)**
HYL3	3	6.3837e-1 (6.69e-3) =	6.3482e-1 (6.16e-3) =	6.3521e-1 (6.39e-3) =	**6.3340e-1 (1.83e-2) +**	6.3477e-1 (7.19e-3)
HYL4	4	9.1961e-1 (7.35e-3) =	**9.1795e-1 (6.66e-3) =**	9.1824e-1 (9.02e-3) =	9.1824e-1 (9.47e-3) =	9.1961e-1 (7.86e-3)
HYL5	2	1.6965e-1 (9.75e-4) =	1.6969e-1 (9.27e-4) =	1.6980e-1 (9.86e-4) =	**1.6961e-1 (7.46e-4) +**	1.7013e-1 (8.17e-4)
+/−/ =		2/5/12	0/8/11	0/14/5	2/12/5	

high, it is difficult for an individual to accurately recognize the distribution of high-quality solutions at the early stage of evolution. On the contrary, if the τ value is set too low, the initial orchestration factor is weak, and the evolutionary process of the individual will show a high degree of randomness. In this paper, different values of τ ($\tau = 0.1, 0.2, 0.3, 0.4, 0.5$) were set for the experiments, and the results are shown in Table 1 Table 2. The results indicate that when $\tau = 0.5$, the algorithm demonstrates poor convergence. However, when $\tau = 0.2$, the algorithm achieves optimal performance.

2) Analysis of the decay rate of the orchestration factor ρ

In the optimization process of the algorithm, the decay rate ρ of the orchestration factor plays an important role, which is used to control the decay strength of the orchestration factor, and thus affects the fitness and evolutionary process of the individuals. Specifically, the setting of ρ determines the degree of influence of historical information on the evolution of the current individual, and either too high or too low a value of ρ will have different effects on the performance of the algorithm. When the ρ value is too high, the decay of the orchestration factor is too fast, leading to a rapid weakening of the individual's influence, which may allow the contribution of the historical high-quality solution to be ignored prematurely. On the contrary, if the value of ρ is too low, the decay rate of the orchestration factor is too slow, and the influence of

Table 7
The Friedman test results of each comparative algorithm on each performance indicator.

	Indicators	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
MMF	IGD	3.375	5.25	3.375	2	5.25	**1.75**
	IGDX	**2.375**	2.625	3.875	3.625	6	2.5
MMMOP	IGD	3.25	5.25	2.95	2.9	5.05	**1.65**
	IGDX	2.45	4.85	3.2	3.2	5.05	**2.25**
HYL	IGD	5.2	3.2	4.2	3.4	3	**2**
	IGDX	3.6	4	5	2.6	4.8	**1**

Table 8
Compare the IGD values of the algorithms on the MMF test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
MMF1	2	3.6736e-3 (1.95e-4) -	3.5921e-3 (2.09e-4) -	3.7698e-3 (3.14e-4) -	3.5145e-3 (1.32e-4) -	4.6410e-3 (4.37e-4) -	**3.2398e-3 (1.68e-4)**
MMF2	2	1.9441e-2 (7.93e-3) -	1.6739e-2 (6.10e-3) -	1.2774e-2 (3.53e-3) +	**1.0187e-2 (2.02e-3)** +	4.3953e-2 (1.85e-2) -	1.3331e-2 (1.98e-3)
MMF3	2	1.5025e-2 (5.49e-3) -	1.6972e-2 (1.18e-2) -	1.1698e-2 (5.14e-3) =	**9.0394e-3 (1.36e-3)** +	3.8801e-2 (1.66e-2) -	1.0179e-2 (1.34e-3)
MMF4	2	3.3725e-3 (1.40e-4) -	4.7924e-3 (2.10e-4) -	3.6473e-3 (1.94e-4) -	**3.2054e-3 (6.43e-4)** =	3.7186e-3 (2.41e-4) -	3.2359e-3 (1.23e-4)
MMF5	2	3.5247e-3 (1.71e-4) -	4.1328e-3 (2.47e-4) -	3.5271e-3 (1.77e-4) -	3.6225e-3 (9.47e-5) -	4.0382e-3 (2.93e-4) -	**3.2311e-3 (1.51e-4)**
MMF6	2	3.4885e-3 (2.06e-4) -	3.9502e-3 (2.77e-4) -	3.5131e-3 (1.94e-4) -	3.5432e-3 (1.26e-4) -	3.8915e-3 (3.13e-4) -	**3.2159e-3 (1.43e-4)**
MMF7	2	3.7127e-3 (2.54e-4) -	5.2103e-3 (3.73e-4) -	4.0490e-3 (2.31e-4) -	3.6643e-3 (4.48e-4) -	3.7553e-3 (2.34e-4) -	**3.1725e-3 (1.40e-4)**
MMF8	2	3.9925e-3 (2.86e-4) -	5.2054e-3 (2.63e-4) -	3.6546e-3 (1.96e-4) +	**2.7546e-3 (3.89e-4)** +	4.6943e-3 (2.59e-4) -	3.8171e-3 (2.21e-4)
+/−/ =		0/8/0	0/8/0	2/5/1	3/4/1	0/8/0	

Table 9
Compare the IGDX values of the algorithms on the MMF test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
MMF1	2	4.0234e-2 (1.38e-3) +	**3.8325e-2 (1.38e-3)** +	4.6002e-2 (2.87e-3) -	4.5234e-2 (1.38e-3) -	6.1264e-2 (6.05e-3) -	4.1941e-2 (1.28e-3)
MMF2	2	4.3138e-2 (1.31e-2) -	3.6229e-2 (1.49e-2) =	3.1853e-2 (8.75e-3) =	**1.8006e-2 (3.97e-3)** +	8.7788e-2 (3.69e-2) -	3.1702e-2 (9.55e-3)
MMF3	2	3.2794e-2 (8.28e-3) -	3.7225e-2 (1.79e-2) -	2.9565e-2 (9.50e-3) -	**1.6582e-2 (3.97e-3)** +	7.7659e-2 (4.31e-2) -	2.3907e-2 (2.30e-3)
MMF4	2	2.0648e-2 (5.98e-4) =	2.3999e-2 (8.31e-4) -	2.5512e-2 (2.14e-3) -	2.7427e-2 (2.61e-3) -	4.1776e-2 (6.56e-3) -	**2.0388e-2 (5.07e-4)**
MMF5	2	6.4378e-2 (2.20e-3) +	**6.3438e-2 (2.00e-3)** +	7.6852e-2 (2.94e-3) -	7.8027e-2 (2.86e-3) -	1.1023e-1 (1.09e-2) -	7.1354e-2 (2.16e-3)
MMF6	2	**5.6436e-2 (1.54e-3)** +	5.8721e-2 (1.31e-3) +	6.5016e-2 (2.47e-3) -	6.7734e-2 (2.31e-3) -	9.3324e-2 (8.19e-3) -	6.0277e-2 (8.67e-4)
MMF7	2	1.9793e-2 (8.93e-4) =	2.3416e-2 (1.31e-3) -	2.5146e-2 (2.60e-3) -	2.7340e-2 (2.35e-3) -	3.8080e-2 (5.64e-3) -	**1.9793e-2 (8.49e-4)**
MMF8	2	**5.0931e-2 (6.01e-3)** +	5.1386e-2 (4.30e-3) +	8.3747e-2 (1.30e-2) =	6.9723e-2 (1.23e-2) +	1.1304e-1 (3.13e-2) -	8.7678e-2 (1.52e-2)
+/−/ =		4/2/2	4/3/1	0/6/2	3/5/0	0/8/0	

the historical factors on the individual is too large, which may easily lead the individual to fall into a state of over-reliance on historical information, and thus lack of adaptability to the current environment. In this paper, different ρ values ($\rho = 0.1, 0.2, 0.3, 0.4, 0.5$) are set for the experiment, and the results are shown in Table 3 Table 4. When $\rho = 0.5$, the decay rate is too fast, leading to poor performance on most problems. However, the best results are achieved when $\rho = 0.1$.

3) Analysis of Parameter k

The parameter k determines the allocation of computational resources between the search phase and other phases of the algorithm. Specifically, k controls the exploration intensity of the population: a higher k value weakens the algorithm's exploratory capability within the solution space, potentially hindering the effective discovery of promising regions and thereby diminishing the ability to locate the global optimum. Conversely, a lower k value increases the algorithm's global search propensity, but may disperse computational resources and delay the population's convergence. Therefore, selecting an appropriate k value is essential for balancing population diversity and enhancing convergence performance. Consequently, in this study, experiments were conducted with varying values of k ($k = 0.1, 0.2, 0.3, 0.4, 0.5$), and the corresponding results are summarized in Tables 5 and 6. The results indicate that when $k = 0.1$, performance is suboptimal due to a reduced

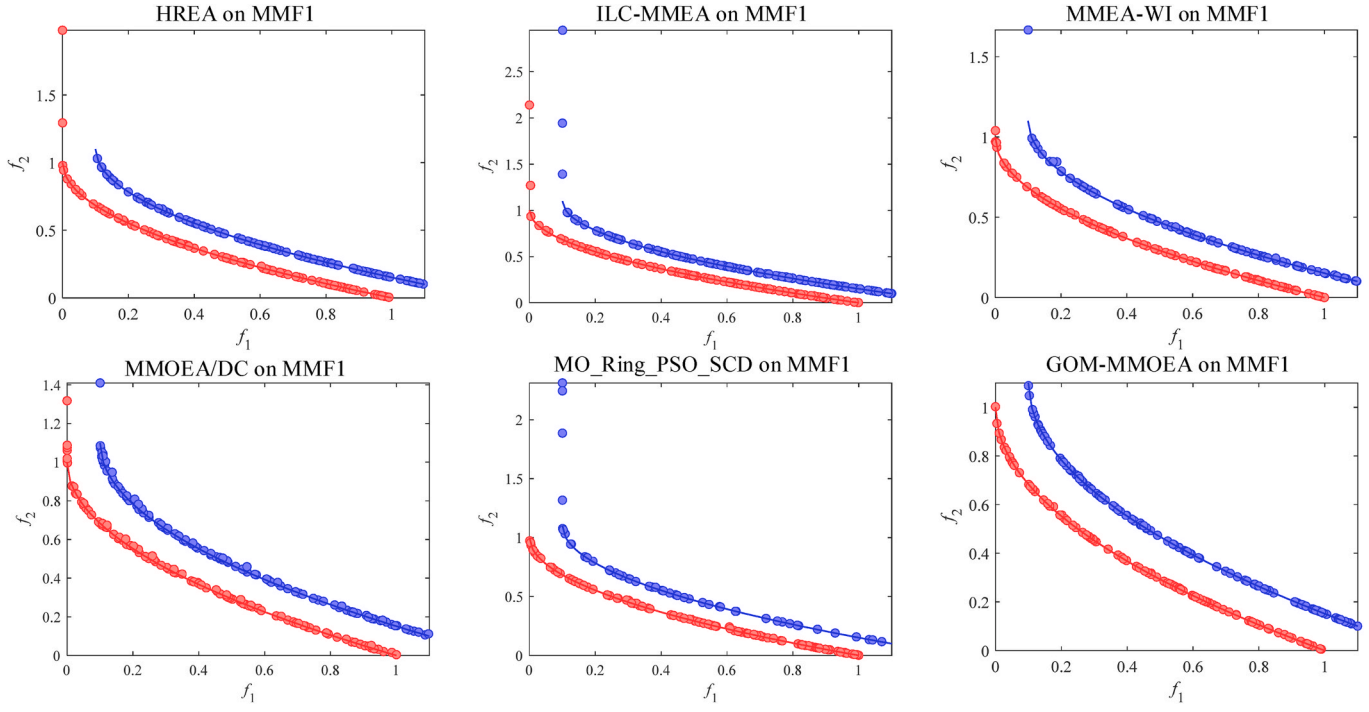


Fig. 3. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMF1 problem.

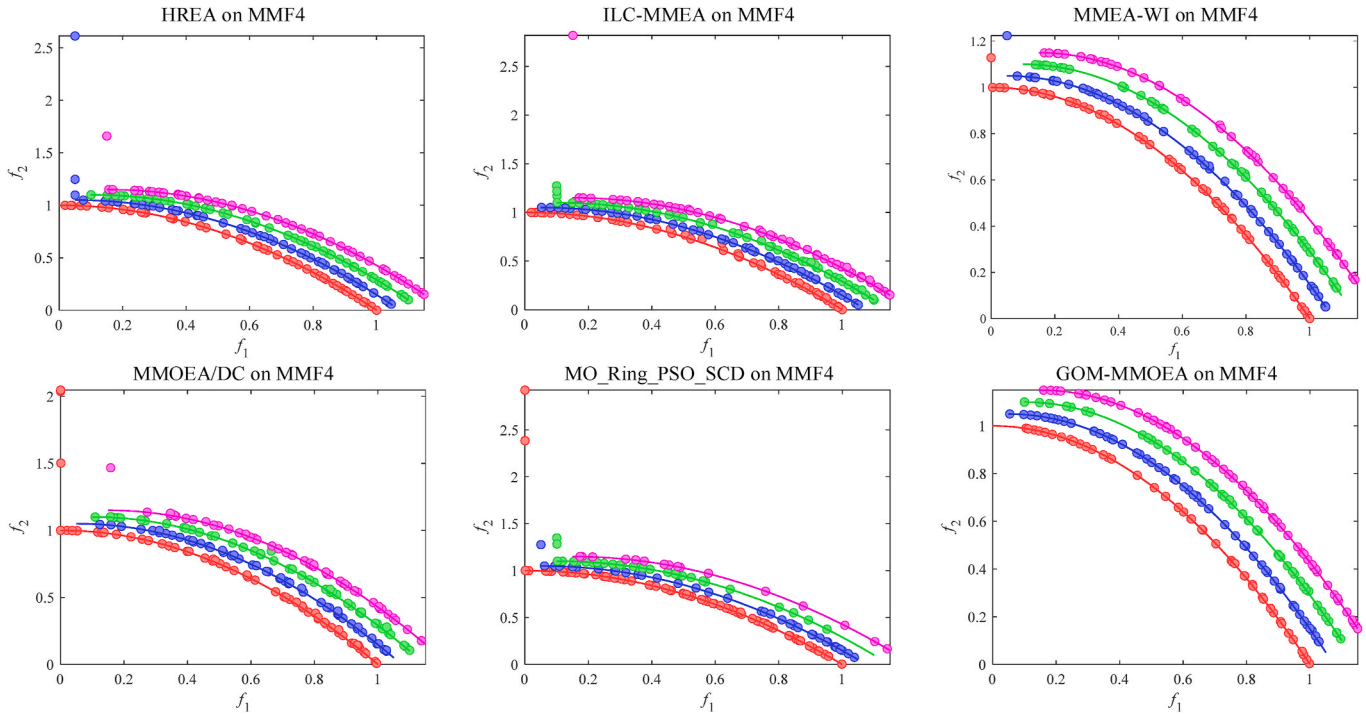


Fig. 4. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMF4 problem.

emphasis on the convergence stage. Conversely, when $k = 0.5$, the algorithm exhibits insufficient diversity. Overall, the optimal balance between convergence and diversity is achieved at $k = 0.4$, yielding the best results.

4.4. Comparison of algorithm performance

This section tests the GOM-MMOEA algorithm and compares it with its counterparts on the MMF, MMMOP, and HYL test problems. Table 7 summarizes the average rankings of six algorithms based on the IGD and IGDX metrics across the three test suites. In the MMF test problem, GOM-MMOEA achieves the best ranking for the IGD metric, followed by MMOEA/DC in second place, while ILC-MMEA and MO_Ring_PSO_SCD perform less favorably. For the IGDX metric, HREA ranks first, with GOM-MMOEA closely following, while MO_Ring_PSO_SCD performs poorly. GOM-MMOEA achieves the best ranking for both the IGD and IGDX metrics on the MMMOP and HYL test problems, demonstrating the stability and efficiency of the GOM-MMOEA algorithm in addressing

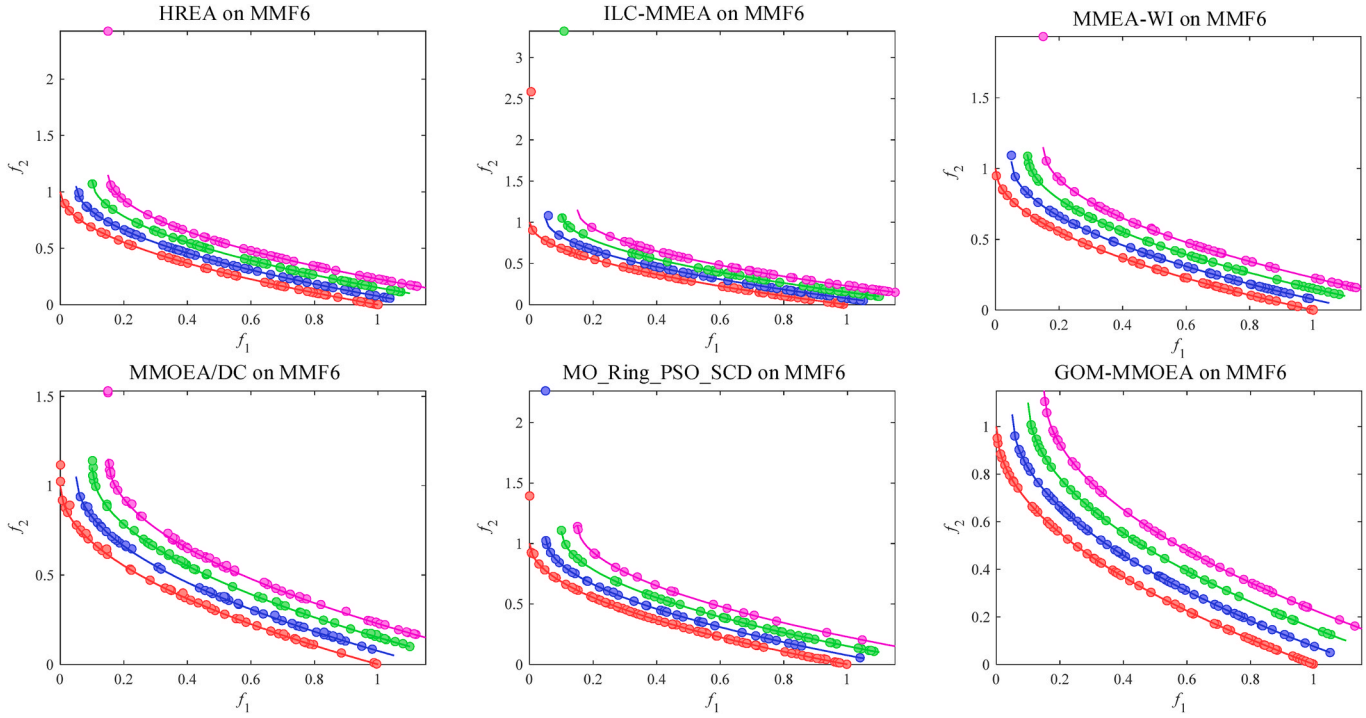


Fig. 5. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMF6 problem.

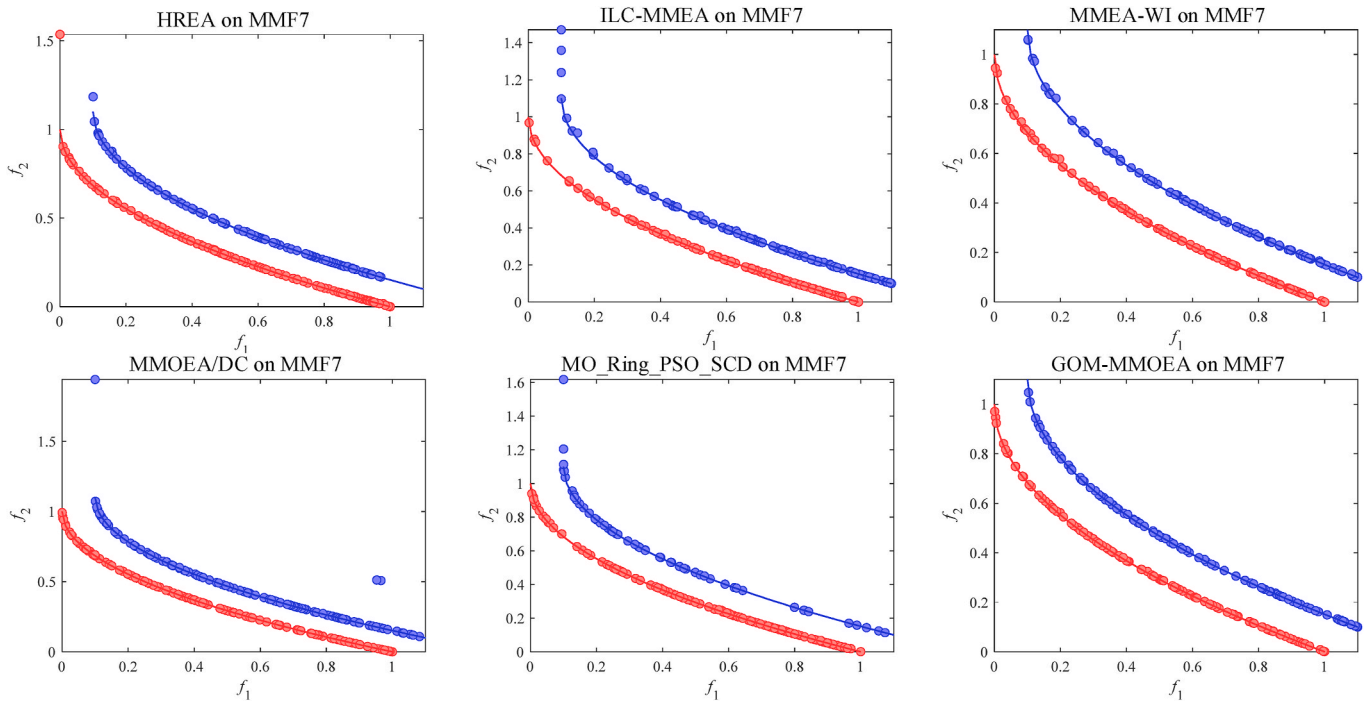


Fig. 6. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMF7 problem.

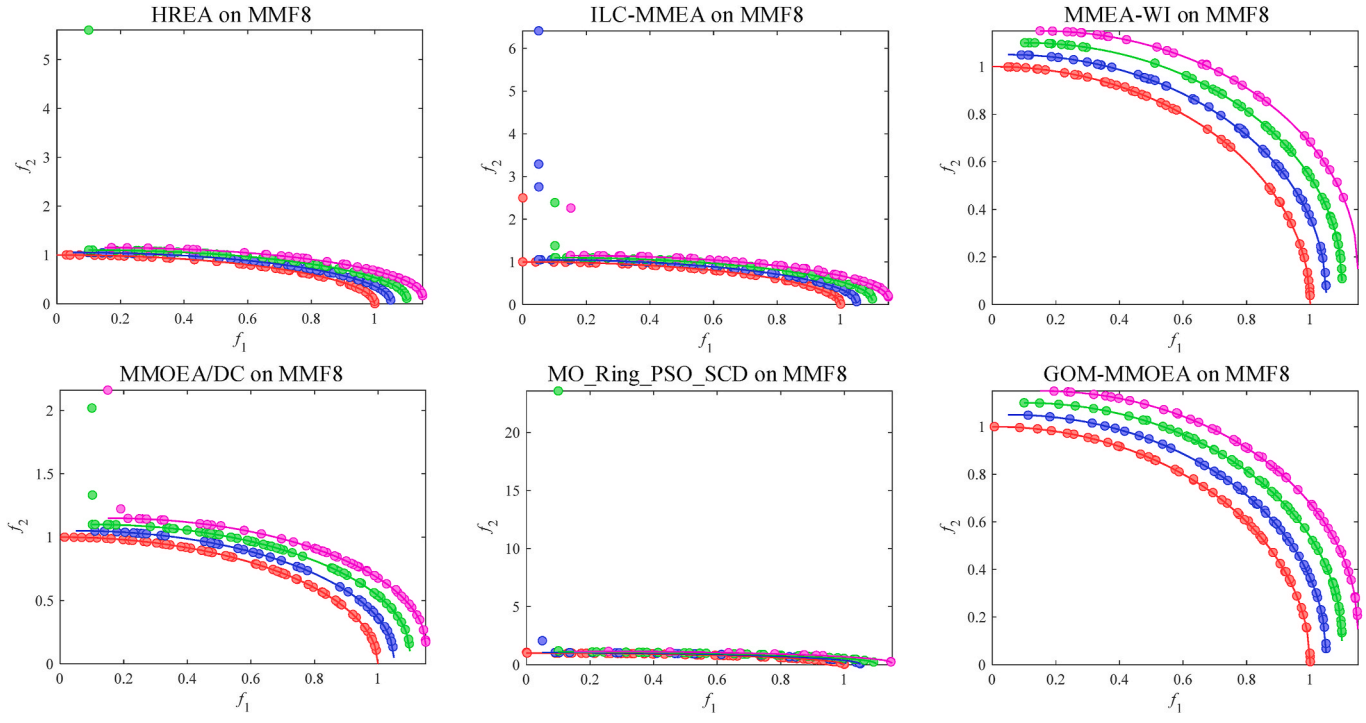


Fig. 7. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMF8 problem.

MOPs.

In the objective space, there are cases where multiple PSs correspond to the same PF. Therefore, to present the experimental results more clearly, in this paper, the real PF are transformed so as to present the PF generated by different PSs. Specifically, each line represents the PF generated by different PSs, respectively, with appropriate translation treatment, and the number of lines reflects the number of PSs corresponding to them. This treatment allows us to more intuitively observe the performance of the algorithm in the decision space versus the objective space.

The IGD and IGDX values for the comparison algorithms and GOM-MMOEA on MMF test problems are presented in Tables 8 and 9. In Table 8, from the IGD results HREA, ILC-MMEA, and MO_Ring_PSO_SCD have higher values on most of the test problems, which shows a lack of performance. MMOEA/DC has higher values on MMF2, MMF3, and MMF8 test problems are more effective and show some advantages. MMEA-WI and MMOEA/DC show better performance on some test problems, but overall performance is average and lacks stability on some

Table 10
Compare the IGD values of the algorithms on the MMMOP test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
MMMOP1_A	2	1.5891e-3 (1.21e-4) -	1.8881e-3 (2.35e-4) -	1.9927e-3 (5.29e-4) -	1.7419e-3 (3.90e-4) -	2.1022e-1 (9.22e-2) -	**1.4609e-3 (1.27e-4)**
MMMOP1_B	3	2.1812e-1 (3.10e-1) -	9.5523e+1 (6.38e+0) -	**5.2266e-2 (2.22e-2)** =	1.5379e+0 (7.04e-1) -	2.9847e+1 (6.53e+0) -	5.8425e-2 (2.63e-2)
MMMOP2_A	2	2.9089e-3 (2.30e-4) -	2.6986e-3 (1.60e-4) -	1.9831e-3 (9.14e-5) +	**1.4019e-3 (7.25e-5)** +	1.6703e-1 (6.91e-2) -	2.3628e-3 (2.23e-4)
MMMOP2_B	3	1.9288e-1 (1.99e-2) -	9.5249e-1 (4.34e-2) -	5.9611e-2 (3.69e-3) =	6.3948e-2 (1.59e-2) =	4.6104e-1 (4.03e-2) -	**5.8786e-2 (3.27e-3)**
MMMOP3_A	2	2.7928e-3 (3.19e-4) -	2.8197e-3 (2.55e-4) -	1.9637e-3 (1.04e-4) =	**1.2263e-3 (1.57e-5)** +	2.0411e-3 (9.28e-5) -	1.9514e-3 (8.57e-5)
MMMOP3_B	3	1.5755e-1 (2.26e-2) -	1.1796e+0 (1.31e-2) -	**3.9035e-2 (1.17e-3)** =	7.1464e-2 (2.32e-3) -	1.1270e-1 (4.18e-3) -	3.9398e-2 (1.28e-3)
MMMOP3_C	2	2.8680e-2 (1.52e-2) -	6.8120e-1 (3.52e-2) -	**2.4158e-3 (1.89e-4)** =	3.3455e-3 (2.67e-4) -	4.9223e-2 (3.98e-3) -	2.4187e-3 (1.68e-4)
MMMOP3_D	3	1.3068e-1 (1.98e-2) -	1.2122e+0 (1.04e-2) -	3.9579e-2 (1.22e-3) -	6.8563e-2 (2.37e-3) -	1.3673e-1 (6.13e-3) -	**3.8641e-2 (8.36e-4)**
MMMOP4_A	2	1.5986e-3 (3.76e-5) +	2.1509e-3 (3.50e-4) -	2.3560e-3 (2.20e-4) -	**1.4318e-3 (3.96e-4)** +	1.1211e-2 (2.12e-3) -	1.6346e-3 (3.67e-5)
MMMOP4_B	3	1.1877e+0 (1.11e+0) -	1.0131e+2 (5.36e+0) -	4.4582e-1 (3.91e-1) =	1.1439e+1 (4.57e+0) -	4.5396e+1 (9.70e+0) -	**3.8467e-1 (1.94e-1)**
MMMOP4_C	2	1.8633e-1 (4.72e-1) -	7.2443e+1 (4.34e+0) -	1.4608e-1 (1.40e-1) =	2.4301e-1 (9.92e-2) -	3.0718e+1 (9.84e+0) -	**7.8140e-2 (4.53e-2)**
MMMOP4_D	3	1.0577e+0 (1.08e+0) -	9.9106e+1 (3.17e+0) -	**1.8145e-1 (1.74e-1)** +	5.3730e+0 (2.42e+0) -	3.6205e+1 (8.64e+0) -	2.2688e-1 (1.28e-1)
MMMOP5_A	2	**1.3867e-3 (2.05e-5)** =	1.9467e-3 (2.67e-4) -	1.8160e-3 (6.79e-5) -	1.6694e-3 (5.60e-4) =	1.1871e-2 (1.90e-3) -	1.3884e-3 (2.29e-5)
MMMOP5_B	3	2.2118e-1 (3.39e-1) =	9.9733e+1 (3.56e+0) -	1.1575e-1 (9.55e-2) -	2.3379e+0 (1.30e+0) -	4.6063e+1 (8.35e+0) -	**6.8967e-2 (2.57e-2)**
MMMOP5_C	2	5.6658e-2 (2.05e-1) -	6.5531e+1 (1.01e+1) -	2.7970e-1 (3.23e-1) -	4.1384e-2 (2.41e-2) =	3.3768e+1 (7.66e+0) -	**3.1042e-2 (1.53e-2)**
MMMOP5_D	3	2.7683e-1 (4.40e-1) -	9.7793e+1 (4.74e+0) -	1.5129e-1 (2.16e-1) -	7.2699e-1 (5.99e-1) -	4.0632e+1 (7.41e+0) -	**6.0603e-2 (2.19e-2)**
MMMOP6_A	2	**1.5346e-3 (4.08e-5)** +	1.8759e-3 (5.34e-5) -	1.7820e-3 (6.17e-5) -	2.0276e-3 (9.44e-5) -	1.9880e-3 (8.32e-5) -	1.5990e-3 (5.19e-5)
MMMOP6_B	3	3.1570e-2 (3.39e-4) -	3.9734e-2 (6.17e-4) -	3.5895e-2 (6.20e-4) -	**3.0405e-2 (5.33e-4)** +	3.4758e-2 (8.80e-4) -	3.0783e-2 (3.41e-4)
MMMOP6_C	2	5.7539e-3 (1.45e-3) -	4.9049e-3 (1.51e-3) -	1.8733e-2 (1.07e-2) -	4.5260e-3 (6.54e-4) -	1.4328e-2 (1.56e-3) -	**4.0128e-3 (9.01e-4)**
MMMOP6_D	3	5.5982e-2 (4.60e-3) -	5.5955e-2 (4.48e-3) -	9.8884e-2 (2.89e-2) -	3.9012e-2 (3.38e-3) -	5.9299e-2 (2.89e-3) -	**3.5860e-2 (2.68e-3)**
+/−/ =		2/16/2	0/20/0	2/11/7	4/13/3	0/20/0	

Table 11
Compare the IGDX values of the algorithms on the MMMOP test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
MMMOP1_A	2	**7.2054e-3 (3.64e-3)** +	3.9815e-2 (2.64e-2) -	6.4167e-2 (3.50e-2) -	8.1871e-3 (6.68e-3) +	2.1416e-1 (4.39e-2) -	2.4888e-2 (1.59e-2)
MMMOP1_B	3	1.7404e-1 (4.47e-2) -	9.8427e-1 (7.07e-3) -	1.8766e-1 (1.73e-2) -	2.6019e-1 (2.73e-2) -	5.0512e-1 (3.64e-2) -	**1.4103e-1 (3.02e-2)**
MMMOP2_A	2	**2.6675e-3 (7.84e-4)** +	3.5577e-3 (1.12e-3) +	3.9833e-3 (4.21e-3) +	3.0003e-3 (2.72e-4) +	1.6627e-1 (8.28e-2) -	9.8121e-3 (3.92e-3)
MMMOP2_B	3	2.2276e-1 (4.29e-2) -	8.7769e-1 (1.14e-2) -	**5.5275e-2 (2.04e-2)** =	1.0244e-1 (3.14e-2) -	6.3997e-1 (1.15e-1) -	5.5444e-2 (6.74e-3)
MMMOP3_A	2	6.1070e-3 (5.18e-4) -	8.6697e-3 (7.00e-4) -	5.3820e-3 (2.78e-4) -	6.2552e-3 (2.49e-4) -	8.5279e-3 (1.00e-3) -	**3.6377e-3 (9.30e-5)**
MMMOP3_B	3	2.7458e-1 (3.41e-2) -	1.0694e+0 (8.09e-3) -	**7.7902e-2 (2.39e-3)** +	2.1350e-1 (1.09e-2) -	2.4331e-1 (1.45e-2) -	8.3514e-2 (1.47e-3)
MMMOP3_C	2	1.0747e-1 (3.69e-2) -	7.9639e-1 (1.28e-2) -	9.9001e-2 (4.28e-2) -	1.0402e-1 (3.78e-2) -	1.9092e-1 (1.95e-2) -	**1.3673e-2 (3.57e-4)**
MMMOP3_D	3	2.5273e-1 (3.01e-2) -	9.7576e-1 (4.40e-3) -	**8.2115e-2 (1.99e-3)** +	2.2224e-1 (8.03e-3) -	2.7622e-1 (1.17e-2) -	8.5090e-2 (9.92e-4)
MMMOP4_A	2	**6.5453e-4 (5.55e-6)** +	8.4365e-4 (1.31e-4) -	1.0451e-3 (4.43e-5) -	2.3281e-3 (6.92e-4) -	1.0742e-2 (3.27e-3) -	6.5681e-4 (1.95e-6)
MMMOP4_B	3	1.9158e-1 (5.86e-2) -	1.0636e+0 (2.96e-2) -	1.9626e-1 (6.36e-2) -	3.3944e-1 (7.16e-2) -	4.6318e-1 (4.37e-2) -	**1.5132e-1 (1.50e-2)**
MMMOP4_C	2	9.0962e-2 (8.42e-2) =	7.8508e-1 (1.16e-2) -	1.6012e-1 (6.70e-2) -	**3.0717e-2 (1.59e-2)** +	4.5031e-1 (5.05e-2) -	5.8265e-2 (1.62e-2)
MMMOP4_D	3	1.8947e-1 (7.32e-2) =	9.7377e-1 (5.80e-3) -	**1.5930e-1 (5.18e-2)** =	2.6792e-1 (4.39e-2) -	4.9217e-1 (3.28e-2) -	1.6367e-1 (1.94e-2)
MMMOP5_A	2	**6.5421e-4 (5.71e-6)** =	9.3682e-4 (1.30e-4) -	1.0737e-3 (4.41e-5) -	1.2207e-3 (8.56e-5) -	1.0634e-2 (5.51e-3) -	6.5609e-4 (1.39e-6)
MMMOP5_B	3	9.7930e-2 (3.77e-2) =	1.0669e+0 (1.30e-2) -	1.4121e-1 (5.35e-2) -	2.1465e-1 (3.80e-2) -	4.5922e-1 (3.71e-2) -	**7.8408e-2 (1.10e-2)**
MMMOP5_C	2	9.1930e-2 (1.05e-1) =	7.6001e-1 (3.16e-2) -	2.7906e-1 (1.03e-1) -	**6.2926e-3 (1.86e-3)** +	4.5549e-1 (5.54e-2) -	2.9276e-2 (1.17e-2)
MMMOP5_D	3	1.1832e-1 (4.32e-2) =	9.7381e-1 (7.39e-3) -	1.8952e-1 (5.92e-2) -	1.7365e-1 (3.56e-2) -	4.7052e-1 (3.19e-2) -	**1.1173e-1 (2.11e-2)**
MMMOP6_A	2	**3.3203e-3 (8.44e-5)** +	4.0886e-3 (7.93e-5) -	4.2571e-3 (1.13e-4) -	5.3667e-3 (1.85e-4) -	7.0440e-3 (7.45e-4) -	3.5256e-3 (8.68e-5)
MMMOP6_B	3	**3.7830e-2 (2.69e-4)** +	5.5660e-2 (1.12e-3) -	4.7011e-2 (1.12e-3) -	5.9597e-2 (1.70e-3) -	5.2990e-2 (2.67e-3) -	3.8752e-2 (2.49e-4)
MMMOP6_C	2	4.5188e-2 (4.02e-3) +	2.9140e-2 (1.38e-2) +	2.1885e-1 (4.38e-2) +	**2.6472e-2 (1.09e-2)** +	1.4000e-1 (2.56e-2) +	2.7694e-1 (4.01e-2)
MMMOP6_D	3	1.4743e-1 (2.91e-2) +	1.2370e-1 (7.50e-3) +	2.6679e-1 (3.94e-2) =	**1.2094e-1 (3.70e-3)** +	1.8681e-1 (1.16e-2) +	2.4718e-1 (4.02e-2)
+/−/ =		7/7/6	3/17/0	4/13/3	6/14/0	2/18/0	

problems. In contrast, GOM-MMOEA shows the most stable performance in most of the test problems, and its IGD values are significantly lower than those of the other algorithms, reflecting its superiority in terms of solution quality and distribution. According to the data in Table 9, GOM-MMOEA is slightly less effective on most MMF test problems. This is mainly since early MMF test problems focused more on the exploration of the objective space and less on the decision space. Nonetheless, GOM-MMOEA still shows performance advantages when dealing with MMOPs.

In Fig. 3, which illustrates the PF for MMF1, it can be observed that most of the PFs generated by the compared algorithms converge toward the true Pareto front. Fig. 4 shows that, for the MMF4 problem, the Pareto fronts generated by MMEA-WI and GOM-MMOEA are uniformly distributed in the objective space, whereas the convergence of other algorithms is clearly inferior to that of MMEA-WI and GOM-MMOEA. In Fig. 5, for the MMF6 problem, GOM-MMOEA demonstrates more significant convergence, with the generated PF evenly distributed across the entire objective space, while most of its comparison algorithms fail to fully converge to the true PF. Some solutions of the comparison algorithms in Fig. 6 for MMF7 do not fully converge to the true PF, whereas the PFs generated by GOM-MMOEA are evenly distributed throughout the objective space and are closer to the true PF. Finally, in Fig. 7, only MMEA-WI and GOM-MMOEA are close to the true Pareto front, reflecting their superior convergence capabilities, while the other

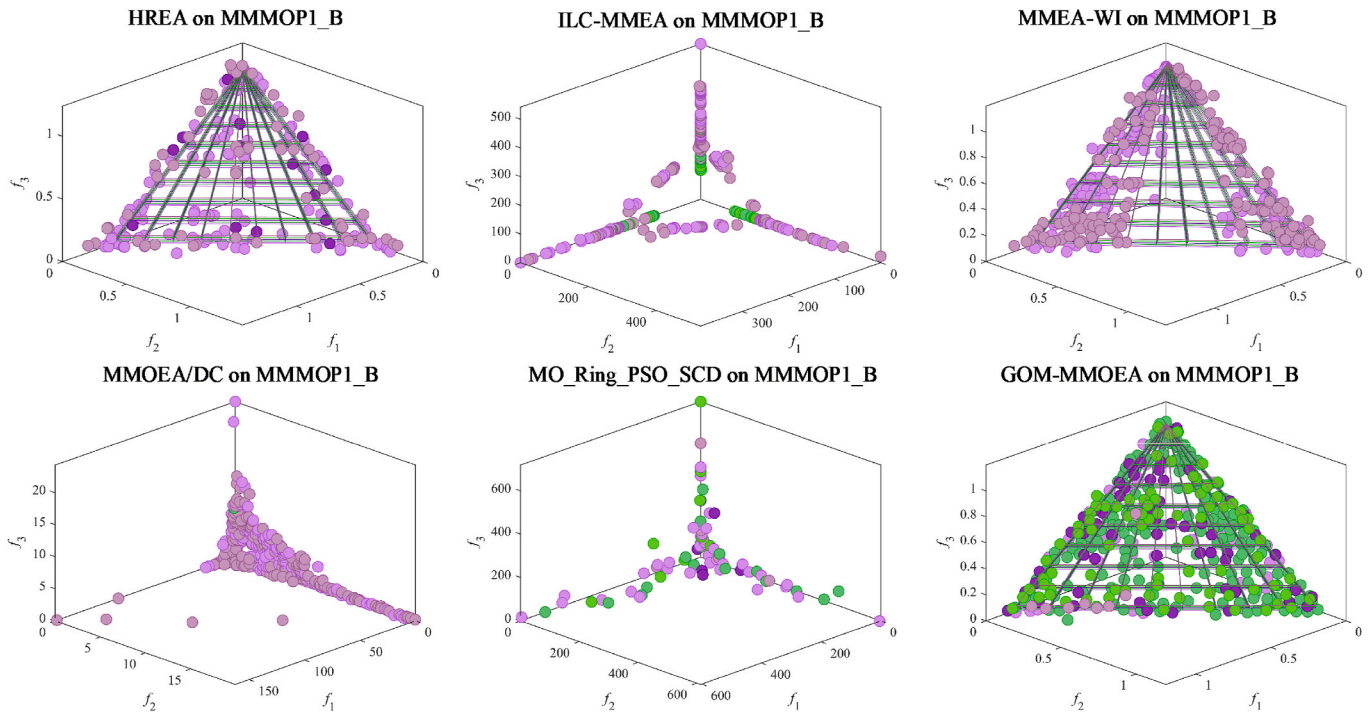


Fig. 8. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMMOP1_B problem.

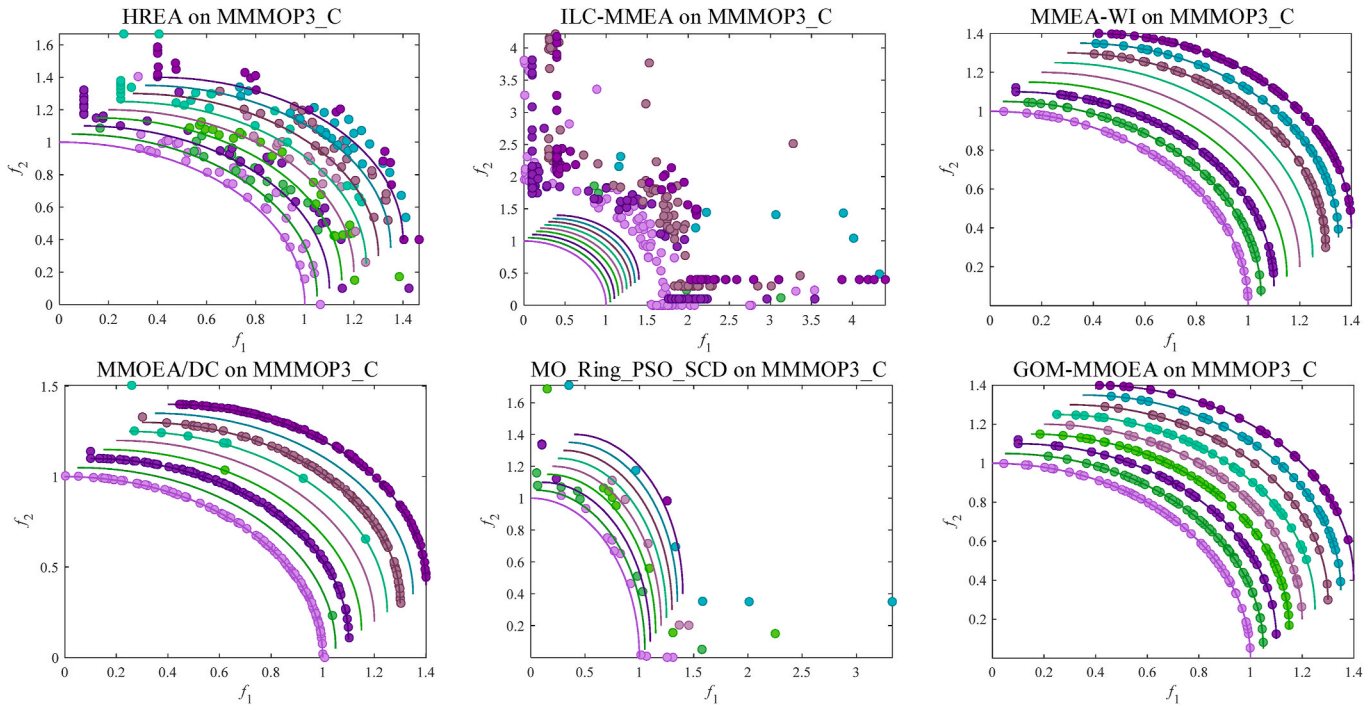


Fig. 9. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMMOP3_C problem.

algorithms fail to fully converge to the true PF in the MMF8 problem.

Table 10 presents the IGD metrics for the six algorithms on the MMMOP test problem. From the results, it is clear that GOM-MMOEA significantly outperforms the other algorithms on 10 test problems and achieves the optimal results. Meanwhile, MMOEA/DC and MMEA-WI also show more excellent performance. Table 11 shows the performance of all algorithms based on the IGDX metrics. GOM-MMOEA achieves the best results on 6 test problems, while the HREA algorithm also shows similar performance.

Figs. 8–10 show the PF of each algorithm in the objective space for the three test problems, providing a more intuitive view of their convergence performance. In the MMMOP1_B test problem, HREA, MMEA-WI, and GOM-MMOEA exhibit better convergence compared to the other algorithms, with GOM-MMOEA closely approximating the optimal Pareto front. Furthermore, GOM-MMOEA demonstrates superior population diversity and a more uniform distribution of solutions than HREA and MMEA-WI. In the MMMOP3_C problem, GOM-MMOEA demonstrates the ability to recognize the complete set of equivalent

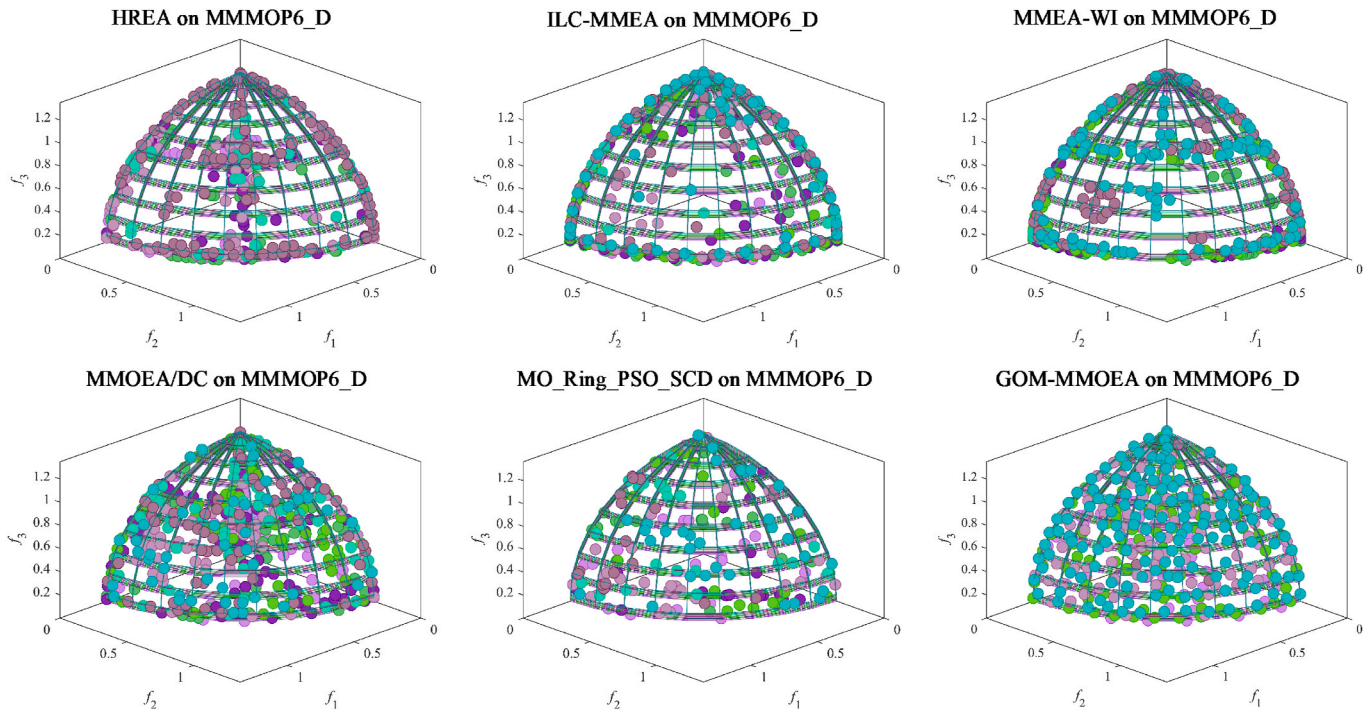


Fig. 10. The Final PF obtained by GOM-MMOEA with each of the compared algorithms on the MMMOP6_D problem.

Table 12
Compare the IGD values of the algorithms on the HYL test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
HYL1	2	2.3126e+0 (1.48e-3) =	2.3126e+0 (3.31e-4) =	2.3222e+0 (1.51e-2) -	2.3121e+0 (6.89e-4) =	**2.3119e+0 (7.25e-5)** =	2.3121e+0 (2.83e-4)
HYL2	2	7.0181e+0 (2.92e-3) -	7.0151e+0 (1.16e-4) =	7.0166e+0 (2.36e-3) =	7.0171e+0 (1.54e-3) -	**7.0150e+0 (8.61e-5)** +	7.0156e+0 (6.17e-4)
HYL3	3	9.0295e+0 (2.89e-2) -	8.9908e+0 (3.99e-3) -	9.0087e+0 (9.43e-3) -	9.0249e+0 (1.24e-2) -	9.0044e+0 (8.08e-3) -	**8.9818e+0 (8.94e-3)**
HYL4	4	9.1042e+0 (4.62e-3) -	9.0362e+0 (1.26e-3) -	8.9693e+0 (8.51e-4) -	8.9735e+0 (1.74e-3) -	9.0176e+0 (5.94e-4) -	**8.7582e+0(3.86e-4)**
HYL5	2	6.1355e+0 (1.99e-4) =	6.1353e+0 (3.77e-4) =	6.1362e+0 (3.34e-4) -	**6.1351e+0 (2.05e-4)** +	6.1365e+0 (5.06e-4) -	6.1355e+0 (1.55e-4)
+/−/ =		0/3/2	0/2/3	0/4/1	1/3/1	1/3/1	

Pareto solutions. Compared with other algorithms, its convergence is particularly remarkable, verifying its strong performance in dealing with complex optimization problems. In the MMMOP6_D problem, MMOEA/DC and GOM-MMOEA show the best performance in terms of population distribution. Not only do they show good convergence, but they also dominate in maintaining a uniform population distribution, a clear contrast to the performance of other algorithms.

Tables 12 and 13 show the performance of each algorithm on the IGD and IGDX metrics on the HYL test problem, respectively. By analyzing the data in the tables, it can be seen that GOM-MMOEA significantly outperforms the other algorithms on the IGDX metrics, which indicates that its distribution of solution sets in the decision space is more uniform and comprehensive. As for the IGD metrics, both GOM-MMOEA and MO_Ring_PSO_SCD show strong performance and are able to generate solution sets closer to the true Pareto front in the objective space.

Figs. 11 and 12 visualize the population distribution within the decision space in the HYL1 and HYL2 problems. As can be seen from the figures, ILC-MMEA, MMOEA/DC, and GOM-MMOEA exhibit better population distribution capabilities on both test problems. It shows that these algorithms can effectively maintain the diversity of the solution set when exploring the decision space, thus covering a wider solution region. Although multiple algorithms show good population distribution ability in both test problems, the population distribution of GOM-MMOEA is obviously more even, which helps to improve the quality and robustness of the solutions, and also highlights the significant advantages of GOM-MMOEA in dealing with complex optimization problems. Through a more balanced population distribution, GOM-MMOEA can effectively avoid falling into local optimal solutions, thus showing higher stability in complex optimization problems.

4.5. Ablation experiment

GOM-MMOEA has two key mechanisms and is effective, which will be analyzed in this section through ablation experiments. The variants of

Table 13
Compare the IGDX values of the algorithms on the HYL test problem.

Problem	M	HREA	I_{LC}-MMEA	MMEA-WI	MMOEA/DC	MO_Ring_PSO_SCD	GOM-MMOEA
HYL1	2	2.1607e-1 (1.37e-2) -	2.0292e-1 (7.43e-4) -	2.4709e-1 (1.02e-2) -	2.0337e-1 (2.22e-3) -	2.2836e-1 (5.46e-3) -	**1.8842e-1 (2.11e-3)**
HYL2	2	4.5657e-1 (1.39e-2) -	4.5291e-1 (2.94e-3) -	6.3180e-1 (5.48e-2) -	4.3754e-1 (5.78e-3) -	5.5648e-1 (3.02e-2) -	**3.8733e-1 (2.32e-3)**
HYL3	3	7.7514e-1 (1.80e-2) -	9.7245e-1 (2.07e-2) -	8.8995e-1 (1.22e-1) -	7.0177e-1 (1.13e-2) -	7.4437e-1 (2.05e-2) -	**6.3477e-1 (7.19e-3)**
HYL4	4	1.1618e+0 (6.74e-2) -	1.2602e+0 (1.68e-2) -	1.1105e+0 (5.56e-2) -	1.0133e+0 (1.44e-2) -	1.2066e+0 (7.98e-2) -	**9.1961e-1 (7.86e-3)**
HYL5	2	1.9643e-1 (5.16e-3) -	1.9910e-1 (1.27e-3) -	2.3233e-1 (9.77e-3) -	2.3129e-1 (7.02e-3) -	2.9209e-1 (1.84e-2) -	**1.7013e-1 (8.17e-4)**
+/−/ =		0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	

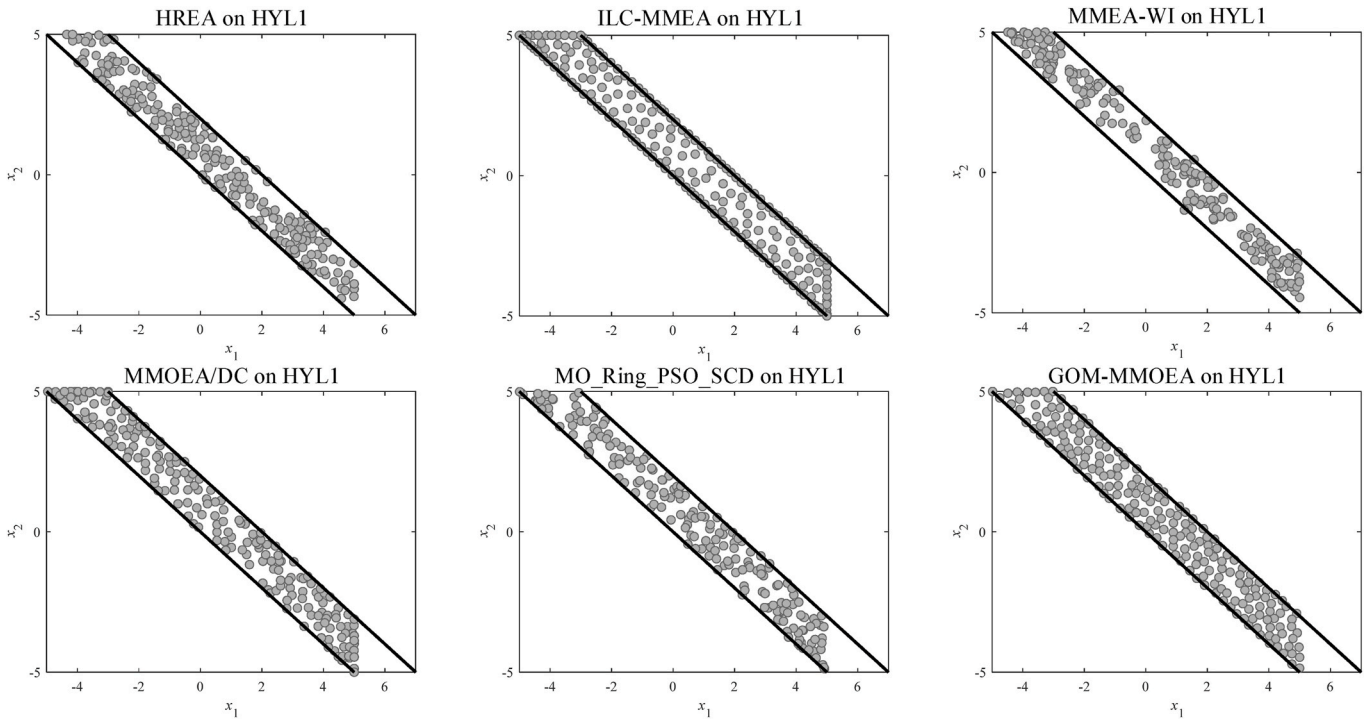


Fig. 11. Reflection of GOM-MMOEA and other comparative algorithms in decision space on the HYL1 problem.

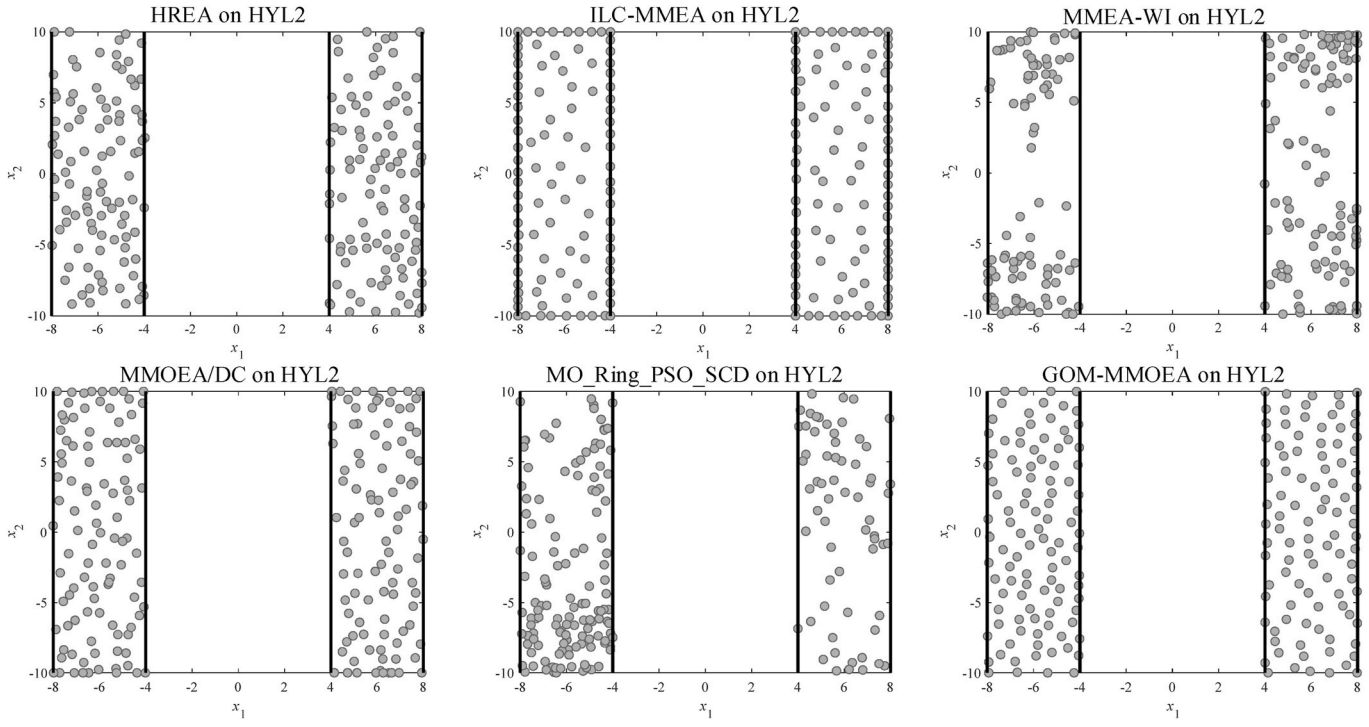


Fig. 12. Reflection of GOM-MMOEA and other comparative algorithms in decision space on the HYL2 problem.

GOM-MMOEA are set up as follows:

GOM-MMOEA-A: In contrast to GOM-MMOEA, this variant does not use the orchestration vector and is explored only on the basis of the fitness values. The fitness value is calculated using the method in SPEA2 (Zitzler et al., 2001) and is mainly used to verify the validity of the orchestration vector.

GOM-MMOEA-B: This variant employs tournament selection instead of the parent selection mechanism proposed in this paper, and is mainly used to validate its effectiveness.

On the MMF test suite, this paper evaluates GOM-MMOEA and its two variants of the algorithm, and the experimental results are shown in Tables 14 and 15. In terms of IGD and IGDX metrics, GOM-MMOEA has a significant advantage in performance, performs optimally in terms of convergence and solution set diversity, and is able to find high-quality PSs efficiently and ensure a good solution set distribution.

On the MMMOP test suite, the experimental results are shown in

Table 14
Compare the IGD values of the algorithms on the MMF test problem.

Problem	M	GOM-MMOEA-A	GOM-MMOEA-B	GOM-MMOEA
MMF1	2	6.1756e-3 (3.90e-4) -	3.3093e-3 (2.07e-4) =	**3.2398e-3 (1.68e-4)**
MMF2	2	2.7045e-2 (2.76e-2) -	1.5645e-2 (3.94e-3) -	**1.3331e-2 (1.98e-3)**
MMF3	2	2.1022e-2 (2.30e-2) -	1.3250e-2 (3.52e-3) -	**1.0179e-2 (1.34e-3)**
MMF4	2	6.4386e-3 (4.67e-4) -	**3.2308e-3 (1.12e-4)** =	3.2359e-3 (1.23e-4)
MMF5	2	6.3574e-3 (4.54e-4) -	**3.1901e-3 (1.34e-4)** =	3.2311e-3 (1.51e-4)
MMF6	2	6.4693e-3 (4.08e-4) -	3.2419e-3 (2.12e-4) =	**3.2159e-3 (1.43e-4)**
MMF7	2	6.6492e-3 (5.03e-4) -	3.2455e-3 (1.48e-4) =	**3.1725e-3 (1.40e-4)**
MMF8	2	8.1680e-3 (7.97e-4) -	3.9872e-3 (3.36e-4) -	**3.8171e-3 (2.21e-4)**
+/−/ =		0/8/0	0/3/5	

Table 15
Compare the IGDX values of the algorithms on the MMF test problem.

Problem	M	GOM-MMOEA-A	GOM-MMOEA-B	GOM-MMOEA
MMF1	2	6.0973e-2 (8.07e-3) -	4.6220e-2 (3.84e-3) -	**4.1941e-2 (1.28e-3)**
MMF2	2	8.7889e-2 (5.50e-2) -	4.1844e-2 (1.91e-2) -	**3.1702e-2 (9.55e-3)**
MMF3	2	5.8387e-2 (3.23e-2) -	3.9825e-2 (1.85e-2) -	**2.3907e-2 (2.30e-3)**
MMF4	2	3.3087e-2 (2.49e-3) -	2.3644e-2 (3.72e-3) -	**2.0388e-2 (5.07e-4)**
MMF5	2	1.0275e-1 (6.64e-3) -	7.9555e-2 (5.30e-3) -	**7.1354e-2 (2.16e-3)**
MMF6	2	8.7586e-2 (5.03e-3) -	6.6989e-2 (4.63e-3) -	**6.0277e-2 (8.67e-4)**
MMF7	2	3.2499e-2 (2.22e-3) -	2.4399e-2 (3.66e-3) -	**1.9793e-2 (8.49e-4)**
MMF8	2	1.4641e-1 (4.29e-2) -	1.6547e-1 (6.87e-2) -	**8.7678e-2 (1.52e-2)**
+/−/ =		0/8/0	0/8/0	

Table 16
Compare the IGD values of the algorithms on the MMMOP test problem.

Problem	M	GOM-MMOEA-A	GOM-MMOEA-B	GOM-MMOEA
MMMOP1_A	2	1.4731e-3 (1.40e-4) =	**1.2680e-3 (1.61e-4)** +	1.4609e-3 (1.27e-4)
MMMOP1_B	3	**2.9679e-2 (3.24e-3)** +	4.8228e-2 (1.55e-2) =	5.8425e-2 (2.63e-2)
MMMOP2_A	2	2.3746e-3 (2.03e-4) =	2.3710e-3 (2.31e-4) =	**2.3628e-3 (2.23e-4)**
MMMOP2_B	3	5.8957e-2 (2.73e-3) =	**5.8093e-2 (2.24e-3)** =	5.8786e-2 (3.27e-3)
MMMOP3_A	2	1.9773e-3 (8.02e-5) =	**1.9294e-3 (9.73e-5)** =	1.9514e-3 (8.57e-5)
MMMOP3_B	3	4.1325e-2 (1.23e-3) -	3.9674e-2 (1.11e-3) =	**3.9398e-2 (1.28e-3)**
MMMOP3_C	2	2.5486e-3 (2.34e-4) -	2.5599e-3 (3.10e-4) =	**2.4187e-3 (1.68e-4)**
MMMOP3_D	3	3.9329e-2 (1.28e-3) -	3.8653e-2 (9.59e-4) =	**3.8641e-2 (8.36e-4)**
MMMOP4_A	2	1.6371e-3 (4.31e-5) =	1.6367e-3 (3.47e-5) =	**1.6346e-3 (3.67e-5)**
MMMOP4_B	3	**7.2737e-2 (4.56e-2)** +	5.5953e-1 (3.67e-1) -	3.8467e-1 (1.94e-1)
MMMOP4_C	2	**4.4964e-3 (2.21e-3)** +	7.6097e-2 (3.91e-2) =	7.8140e-2 (4.53e-2)
MMMOP4_D	3	**4.3164e-2 (8.14e-3)** +	3.0748e-1 (1.48e-1) -	2.2688e-1 (1.28e-1)
MMMOP5_A	2	1.3938e-3 (2.13e-5) =	1.3946e-3 (2.17e-5) =	**1.3884e-3 (2.29e-5)**
MMMOP5_B	3	**3.7014e-2 (5.08e-3)** +	1.4120e-1 (7.78e-2) -	6.8967e-2 (2.57e-2)
MMMOP5_C	2	**3.6111e-3 (1.35e-3)** +	2.9211e-2 (1.71e-2) =	3.1042e-2 (1.53e-2)
MMMOP5_D	3	**3.7081e-2 (8.01e-3)** +	5.9088e-2 (3.29e-2) =	6.0603e-2 (2.19e-2)
MMMOP6_A	2	1.6011e-3 (3.67e-5) =	1.6182e-3 (7.55e-5) =	**1.5990e-3 (5.19e-5)**
MMMOP6_B	3	3.0885e-2 (3.21e-4) =	3.0799e-2 (3.09e-4) =	**3.0783e-2 (3.41e-4)**
MMMOP6_C	2	4.7616e-3 (6.52e-4) -	**3.7290e-3 (1.57e-3)** +	4.0128e-3 (9.01e-4)
MMMOP6_D	3	3.6217e-2 (2.80e-3) =	3.5928e-2 (2.58e-3) =	**3.5860e-2 (2.68e-3)**
+/−/ =		7/4/9	2/3/15	

Tables 16 and 17, which fully highlight the important role of each mechanism in multimodal multi-objective optimization. The IGD metrics in Table 16 show that the GOM-MMOEA-A algorithm, which lacks the orchestration vector, performs poorly in approximating the PF, suggesting that the orchestration vector has a significant advantage in improving the optimization efficiency of the algorithm and avoiding premature convergence. The IGDX data in Table 17 further shows that GOM-MMOEA exhibits optimal results, a result that fully validates the key role of the parent selection mechanism proposed in this paper in the global exploration process.

These experimental results clearly show that GOM-MMOEA achieves the best results, proving the effectiveness of the mechanisms in GOM-MMOEA.

4.6. Computational complexity analysis

Since all algorithms are evaluated on the same benchmark problems, the number of objectives and the dimensionality of the decision space are consistent across comparisons. Therefore, the population size N is used as the primary parameter to assess computational complexity. In GOM-MMOEA, the fitness evaluation involves calculating both the orchestration factor and a density metric for each individual. The density metric is based on the Euclidean distance to the k th nearest neighbor, resulting in a worst-case complexity of $O(N^2)$ due to pairwise distance computations.

Additionally, during each generation, the orchestration vector is dynamically updated based on dominance relations, which also requires comparing each individual with all others, incurring a complexity of $O(N^2)$. The parent selection mechanism further integrates the orchestration factor and crowding distance information, requiring normalization and probabilistic sampling over the entire population, which collectively contributes an additional $O(N)$ cost. During the environmental selection phase, the non-dominated sorting and crowding distance calculation are performed over both parent and offspring populations, with a combined size of $2N$, leading to a complexity of $O(MN^2)$, where MMM is the number of objectives. As the algorithm employs three co-evolving populations that share similar evolutionary operations, the worst-case overall time complexity remains in the order of $O(N^2)$.

In terms of space complexity, each individual stores decision variables, objective values, orchestration factors, and density information. This requires $O(N(D+M))$ memory, where D is the number of decision variables and MMM is the number of objectives. The orchestration vector adds another $O(N)$ space, while pairwise distances for density calculation and dominance comparisons can require up to $O(N^2)$ space in the worst case. Considering the maintenance of three distinct populations, the total space complexity of the algorithm is $O(N(D+M))+O(N^2)$. Given that D and MMM are fixed and relatively small in most

Table 17
Compare the IGDX values of the algorithms on the MMMOP test problem.

Problem	M	GOM-MMOEA-A	GOM-MMOEA-B	GOM-MMOEA
MMMOP1_A	2	1.2066e-1 (8.67e-2) -	9.5679e-2 (8.28e-2) -	**2.4888e-2 (1.59e-2)**
MMMOP1_B	3	1.9371e-1 (8.25e-2) -	1.9400e-1 (6.80e-2) -	**1.4103e-1 (3.02e-2)**
MMMOP2_A	2	4.1504e-2 (4.33e-2) -	4.3622e-2 (5.24e-2) -	**9.8121e-3 (3.92e-3)**
MMMOP2_B	3	1.1853e-1 (9.79e-2) -	1.3901e-1 (1.17e-1) -	**5.5444e-2 (6.74e-3)**
MMMOP3_A	2	3.8278e-3 (1.23e-4) -	3.8758e-3 (1.70e-4) -	**3.6377e-3 (9.30e-5)**
MMMOP3_B	3	8.5990e-2 (1.88e-3) -	8.5193e-2 (2.21e-3) -	**8.3514e-2 (1.47e-3)**
MMMOP3_C	2	2.7578e-2 (3.65e-2) =	2.3175e-2 (3.32e-2) =	**1.3673e-2 (3.57e-4)**
MMMOP3_D	3	8.5966e-2 (1.66e-3) -	8.6408e-2 (1.95e-3) -	**8.5090e-2 (9.92e-4)**
MMMOP4_A	2	6.5803e-4 (3.12e-6) =	6.5920e-4 (2.99e-6) -	**6.5681e-4 (1.95e-6)**
MMMOP4_B	3	**7.4804e-2 (3.99e-2)** +	1.8580e-1 (3.35e-2) -	1.5132e-1 (1.50e-2)
MMMOP4_C	2	2.2932e-1 (8.25e-2) -	1.0443e-1 (6.02e-2) -	**5.8265e-2 (1.62e-2)**
MMMOP4_D	3	**1.2668e-1 (9.13e-2)** +	2.1305e-1 (2.33e-2) -	1.6367e-1 (1.94e-2)
MMMOP5_A	2	6.5711e-4 (2.39e-6) =	6.5762e-4 (2.57e-6) -	**6.5609e-4 (1.39e-6)**
MMMOP5_B	3	**2.8505e-2 (1.47e-2)** +	1.1099e-1 (3.63e-2) -	7.8408e-2 (1.10e-2)
MMMOP5_C	2	2.1643e-1 (8.32e-2) -	1.1886e-1 (9.30e-2) -	**2.9276e-2 (1.17e-2)**
MMMOP5_D	3	1.5451e-1 (1.07e-1) =	1.8220e-1 (6.40e-2) -	**1.1173e-1 (2.11e-2)**
MMMOP6_A	2	**3.4574e-3 (2.18e-4)** +	3.7976e-3 (3.91e-4) -	3.5256e-3 (8.68e-5)
MMMOP6_B	3	3.8836e-2 (3.92e-4) =	3.9453e-2 (5.55e-4) -	**3.8752e-2 (2.49e-4)**
MMMOP6_C	2	3.7091e-1 (5.94e-2) -	3.3428e-1 (7.78e-2) -	**2.7694e-1 (4.01e-2)**
MMMOP6_D	3	3.0976e-1 (7.21e-2) -	3.3335e-1 (6.53e-2) -	**2.4718e-1 (4.02e-2)**
+/−/ =		4/11/5	0/19/1	

scenarios, the overall space complexity is dominated by $O(N^2)$.

In summary, the total computational complexity of GOM-MMOEA is $O(N^2)$. Therefore, GOM-MMOEA achieves improved search performance and population diversity without increasing the computational burden.

5. Conclusions

In this paper, a multimodal multi-objective evolutionary algorithm GOM-MMOEA with global orchestration mechanism is proposed to address the challenges in MMOPs. By constructing an orchestration vector and effectively guiding the distribution of solutions throughout the search process, the algorithm significantly enhances both the exploration ability and convergence performance toward the global optimal solution.

Through a series of benchmark tests, the algorithm is not only able to effectively search for optimal solutions of multiple objectives, but also maintains the diversity of solutions, and the present algorithm achieves remarkable results on several MMOPs. The experimental results show that the algorithm outperforms most other optimization algorithms in terms of uniformity and diversity.

Although the present algorithm performs well in MMOPs, there are still several areas for optimization and improvement. Potential directions for future research include, but are not limited to, the following: (1) The current algorithm primarily relies on global search strategies, and its local search capability may be insufficient. A local search mechanism could be introduced to strengthen the optimization of local solutions during the population convergence process. (2) The current orchestration mechanism reflects the quality of solutions through a fixed update strategy. However, the characteristics of different problems may require a more flexible orchestration approach. Consideration could be given to introducing an adaptive orchestration mechanism, which would allow for more precise control over the balance between convergence and exploration.

CRediT authorship contribution statement

Shaobo Deng: Writing – review & editing, Validation, Supervision. **Hui Shi:** Writing – original draft, Software, Methodology, Investigation. **Hangyu Liu:** Data curation, Conceptualization. **Jinyu Xu:** Resources, Project administration. **Sujie Guan:** Visualization. **Min Li:** Funding acquisition, Formal analysis. **Zhuolei Duan:** Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Research on this work is partially supported by grants from the Jiangxi Province Science and Technology, PR China (No. 20224BAB202014); the National Natural Science Foundation of China under Grant (No. 62166028).

Data availability

Data will be made available on request.

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